



**STUDY ON THE MICROSTRUCTURE AND
MECHANICAL PROPERTIES OF DISSIMILAR
METAL JOINTS OF INCONEL 625 AND
STAINLESS STEEL 316L**



A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report “**STUDY ON THE MICROSTRUCTURE AND MECHANICAL PROPERTIES OF DISSIMILAR METAL JOINTS OF INCONEL 625 AND STAINLESS STEEL 316L**” is the bonafide work of “**PRANAV.K (91001038), SRIRAM.A (191001058), SUDHARSHAN.R (191001059)**” who carried out the project work under our supervision.

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ABSTRACT

Among cutting-edge methods, laser welding is considered a flexible rapidly developing technique with numerous engineering industry applications. It is capable of welding narrow and difficult joints precisely, which can be operated through computer laser welding instruments used for joining different materials including titanium, aluminum, and magnesium alloys and ceramics

Inconel and Stainless-steel weldments are widely used in structural industries. In this study, the dissimilar joints of SS316L to IN625 are made with Fibre Laser beam welding processes. This combination of materials is of interest in chemical plants, oil & gas, and nuclear applications, where high corrosion and wear resistance are essential requirements.

The parameters of the welding process are selected using Taguchi L9 method and its impact on the laser welding quality is studied. The parameters studied were welding speed and laser power. Tensile strength and Hardness profile are measured for the weldment to assess the quality of the welds. Microstructural analysis was done using optical microscopy. The mechanical properties and microstructural characteristics were analysed.

The results show that full penetration for dissimilar weldment takes place at a lower speed of the weld. Formation of brittle phases on the stainless steel side welds at high speeds. The influence of the laser welding parameters was studied and calculated using Anova software.

Keywords: *Laser Welding, dissimilar welding, microstructure, Mechanical testing,*

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbols

mm	-	Millimeter
CM	-	Centimeter
S	-	Second
MPa	-	Megapascal
KPa	-	Kilopascal
HV	-	Vickers Hardness
KN	-	Kilo Newton
kW	-	Kilo Watt

Abbreviations

TS	-	Test sample
LBW	-	Laser Beam Welding
ANOVA	-	Analysis of variance
HAZ	-	Heat affected Zone
AISI	-	American Iron and steel institute
UTS	-	Ultimate Tensile Strength
DOE	-	Design Of Experiments

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO WELDING

Welding is a fundamental and versatile fabrication process that plays a crucial role in the manufacturing, construction, automotive, and aerospace industries, among others. It is the art and science of joining two or more pieces of metal or thermoplastic materials together to create a strong and permanent bond. By melting and fusing the materials, welding creates a continuous connection and that allows for the transmission of forces and the integrity of the structure. Unlike methods that rely on additional components, welding creates strong and permanent bonds by melting and fusing the base materials together. This unique characteristic of welding provides several key advantages.

Versatility is another notable advantage of welding. It can be used to join a wide range of materials, including metals and thermoplastics, enabling the fabrication of diverse components tailored to specific needs. This flexibility makes welding suitable for various applications and industries, from automotive and aerospace to construction and manufacturing. Efficiency and cost-effectiveness are also key advantages of welding over traditional joining processes. Welding eliminates the need for additional fasteners, reducing material and labour costs. It enables the efficient assembly of large structures and components, contributing to improved productivity and reduced manufacturing time. Additionally, welding allows for high production rates, making it a cost-effective choice for mass production compared to the time-consuming process of installing individual nuts and bolts or rivets.

1.2 CLASSIFICATION OF WELDING PROCESSES

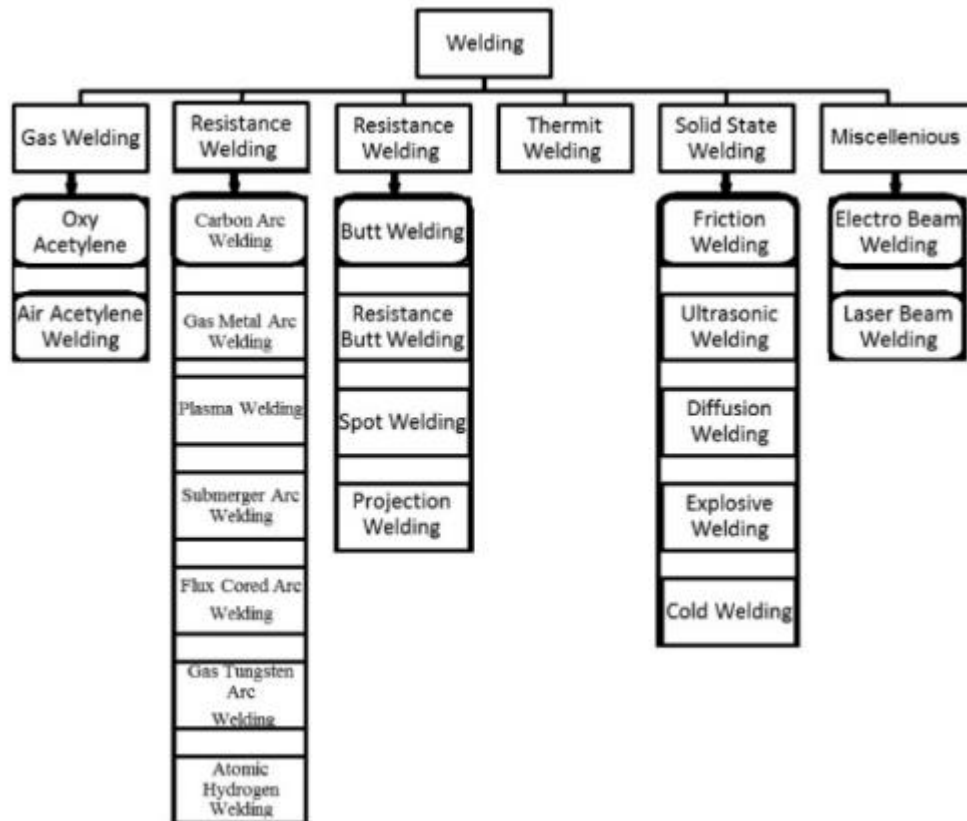


Fig1.1: Classification of welding processes

Welding can be broadly classified into three major groups.

1. Fusion Welding

In the process of fusion welding, the base metal is melted by the application of heat. In order to facilitate this process and provide bulk and strength to the joint, a filler material is used. The following are the most common types of fusion welding processes:

* Arc Welding

- * Resistance Welding

- * Oxyfuel Welding

2. Solid-state Welding

Through solid-state welding, the application of pressure often times accompanied by heat aids in the joining of parts. The salient feature of this type of welding process is that no filler material is used.

The following are the most common types of solid-state welding processes:

- * Diffusion Welding

- * Friction Welding

- * Ultrasound Welding

3. High Precision Welding Processes

These welding techniques are a sub-section of fusion welding processes. They are employed when the result to be achieved must be of very high precision. Welding takes place through the application of heat, wherein the heat supplied may be of varied sources as opposed to conventional welding techniques. The following are the most common types of high precision welding processes:

- * Electron Beam Welding

- * Laser Beam Welding

1.3 LASER WELDING

Laser welding is an advanced welding technique that utilizes a high intensity laser beam as a heat source to join materials together. It offers several advantages over conventional welding methods, making it a preferred choice in various industries such as petrochemical, aerospace industries, etc. In laser welding, a laser beam is directed onto the joint area of the materials to be welded. The laser beam moves along the joint, the molten material solidifies, creating a strong and durable weld joint. The application of laser welding has received a great attention in past few decades due to its higher productivity, higher speeds, and greater penetration as compared to other welding methods with an exception of electron beam welding.[1]

1.4 LASER WELDING OPERATION

Laser beam welding produces small heat-affected zones and a rapid pace of heating and cooling due to its high power density of about $1\text{MW}/\text{cm}^2$. Although the spot size of the laser beam used for welding can range from 0.2mm to 13mm, the smaller size of the laser spot is preferable for the majority of applications. The amount of power applied determines the depth of penetration, which is also influenced by the focus point. Penetration is greatest when the focal point is just below the workpiece's surface. The employment of the laser in continuous mode for deep weld applications will depend on the requirements of the application. Due to the provision of the focal systems, the laser beam can be shaped easily. Thus laser welding can be performed in two main ways, Keyhole type, and Conduction type.

1.4.1 KEYHOLE LASER WELDING

During keyhole laser welding, a focused laser beam is directed onto the workpiece, creating a localized heat source. The laser power and beam characteristics are carefully controlled to generate a high-energy density at the target area. As the laser beam interacts with the material, it rapidly heats and vaporizes the surface layers. The high energy of the laser beam creates a cavity or "keyhole" within the material. The keyhole is formed due to the intense vaporization of the material and the generation of a vapor-filled cavity. The laser power and welding speed are adjusted to maintain the keyhole stability and prevent its collapse. As the laser beam passes through the keyhole, it melts and fuses the material walls surrounding the keyhole. This results in deep penetration welding, as the molten material fills the keyhole and solidifies, forming a strong weld joint. The keyhole acts as a channel for the laser beam, allowing deep penetration into the material, typically exceeding the weld depth achievable with conventional welding methods. Keyhole laser welding offers several advantages, including high welding speeds, narrow heat-affected zone, minimal distortion, and strong weld quality. It is commonly used in various industries, such as automotive, aerospace, electronics, and medical, where precision and high-quality welds are crucial.

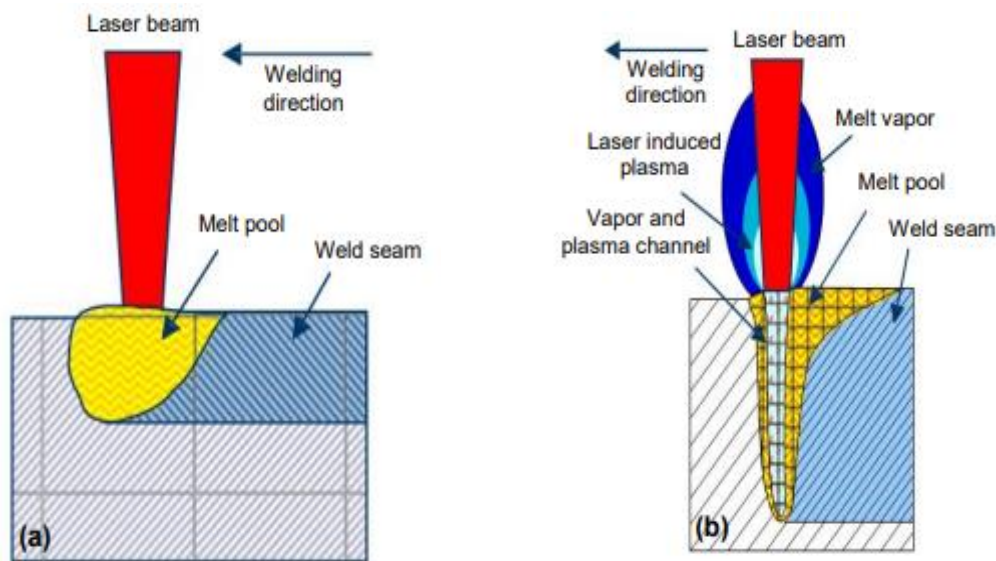


Fig1.2:a)Conduction welding , b)Keyhole welding

1.4.2 CONDUCTION LASER WELDING

The key difference between keyhole and conduction welding modes is the amount of power density employed to generate heat, which is typically influenced by elements like productivity, stability, spatter, etc. Keyhole welding is characterized by deep penetration, small heat affected zone and good productivity. But there are also certain issues with it, namely high levels porosity , instabilities in energy control on the welding surface, spatters and some other weld defects. Most of the above defects occur during welding materials, especially metals with low thickness and high lustre like aluminium and its alloys , steels , etc. There is also the compulsion of use of good laser beam quality for keyhole welding which increases the operation costs. As the power density used is low for conduction welding , this process becomes a stable one as there is much control of the heat supplied to the material. Unlike keyhole welding , this process does not compulsorily require use of high quality laser beam which brings the overall operation costs.

1.5 MODES OF OPERATION OF LASER WELDING

On the basis of operation modes, laser welding can be divided into pulsed laser welding and continuous laser welding. For pulsed laser welding, the range of pulse durations is 1–20 milliseconds, with lower-end pulse frequencies reaching 1 kHz. The only difference between the two modes of operation is that the time needed will vary depending on the mode. Both modes of operation can be used with the same power rating. The work material, weld size and thickness, material qualities, weld bead size, and application factors all play a role in the choice of laser welding operation mode.

1.6 INDUSTRIAL APPLICATIONS OF LASER WELDING

One of the most recent and cutting-edge joining methods being used in a variety of industries for component production as an alternative to older methods like gas and arc welding is laser welding.. Welding with conventional methods of gas and arc welding became an issue with increasing

Technological development, enhanced materials and complex applications .Laser welding was embraced because it provided a number of advantages like acceptable aesthetic appearance, low weld distortion ,reduced material wastage, high production rates ,reduced component size, easy automation , high weld quality , etc. Thus, laser welding has been expanding its horizon in the joining industries as many examples stated can support that.

Laser welding is applied in welding of stainless steel components for the television gun where previous techniques like riveting, soldering and micro resistance spot welding were applied but were not that productive. Laser welding is used for spot welding applications in floating blade cartridges of

razon blades where the size of blades is very important factor. Fixing of opto-electronic transmitter for accurate positioning of thin optical fiber laser welding instead of soldering process or application of adhesives because of its low distortion of workpiece. Laser welding is being applied for joining passenger car body parts because it satisfies the conditions required by the automotive industry for weld and after weld like sufficient ductility for further blanking and pressing operations, narrow weld , minimum heat affected zone, flushing of weld beads with sheet surface to avoid blanking and other press tool damages , in comparison to TIG welding. Laser welding is used in other applications like joining of stainless steel finned tubes to walls of heat exchanger for effective conduction, joining of diamond or carbide impregnated steel tips to rotary machine tools saw blades, Joining of viscous couplings and differentials in four wheel drive systems.[][]

1.6.1 LASER WELDING OF STAINLESS STEEL

When it comes to laser welding, stainless steels offer the highest weldability of any steel. High quality weld can be achieved through the entire range of stainless steels with weld quality decreasing over the order stated; austenitic ,ferritic and martensitic stainless steels. Austenitic steels have the thermal conductivity one third of the carbon steels and 50% greater thermal expansion of the carbon steels ,thus making it a good laser light absorber and resulting in increased laser penetration. The high heat-cooling rates of the process are also an additional bonus to the joining technique as when these steels which does not occur in laser welding case. Martensitic stainless steels have the lowest weldability among the stainless steels for the laser welding

process due to the presence of high carbon content. Martensitic microstructure is brittle in nature and always leads to solidification cracking.[]

1.6.2 ADVANTAGES OF LASER WELDING

Laser welding, nowadays, is increasingly being used in industrial production ranging from microelectronics to shipbuilding. Automotive manufacturing, however, is among the industrial sectors which have proven to be most outstanding at developing applications that take advantage of the many benefits of this technology:

- Low heat input;
- Small heat affected zone (HAZ);
- Low distortion rate; and
- High welding speed.

These characteristics have made laser welding the process of choice for many applications that used conventional welding in the past. By adding the benefits of single-sided access, laser welding is given another strategic advantage, allowing it to open the door for a multitude of new applications.

Modern CNC multi-axis laser welding workstations can provide precise and repeatable welding. A successful laser weld, however, requires a combination of appropriate weld joint design, motion control, fixture design, metallurgy, production control, and quality systems. All three of the most common joint configurations can be used: lap, butt, or fillet, but intimate contact between the materials at the joint is preferred. Under these conditions, high quality, low

porosity, and controlled grain size welds with consistent weld depth and width can be achieved. Weldable materials range from normal and high-strength constructional steels through to high-alloyed stainless steels. Titanium, aluminum, and nickel base materials can also be problem-free welded. Besides, high heating and cooling rates produce fine microstructures, which improve mechanical properties of the weld. Since the heat source is the energy of light, the workpiece is welded purely resulting in excellent fatigue strength of the welded joint as well.

The laser welds can be cost competitive based on minimum set-up time, low fixturing costs, high feed rates, and high-energy efficiency. Resultant welds generally have minimum part distortion with reduced rectification and decreased post-weld costs.

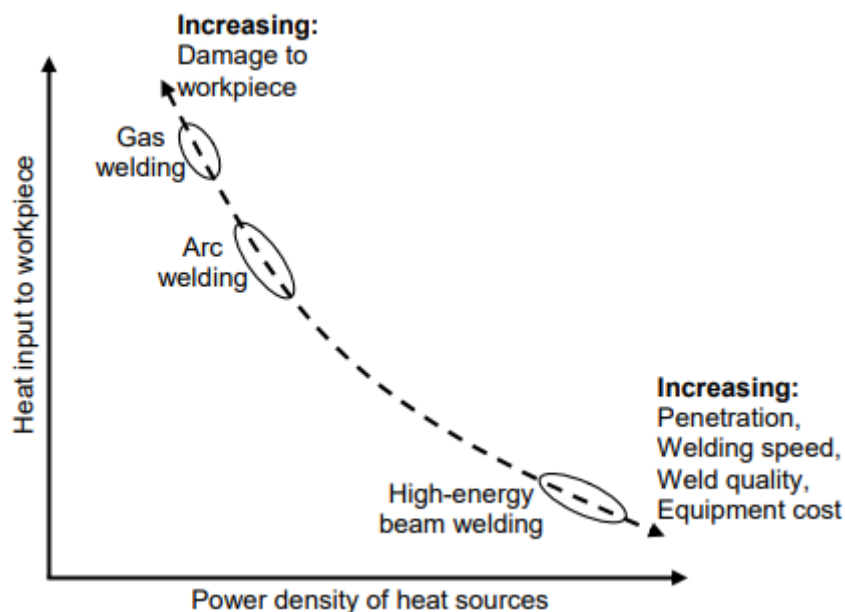


Fig1.3:Power density vs Heat input of different welding techniques

1.7 RESEARCH MOTIVATION

Various studies have been reported on the welding of Different grades of Inconel and stainless steel dissimilar metals using conventional welding techniques. However, studies on laser welded joints of Inconel and stainless steels have not been extensively studied. Specifically greatly used combination of Inconel 625 and stainless steel 316l was not reported in early studies. The advantages of laser welding on austenitic stainless steel compared to other welding techniques, low and controlled heat input and low distortion of metals formed the motivation of the present work which aims at improving the weldability of Inconel 625 and stainless 316l using laser welding

1.7.1 UNIQUENESS OF THE PRESENT WORK

The uniqueness of the present research work is that it aims at improving the weldability of the Inconel 625 and stainless steel 316l dissimilar joints using the latest high energy density laser welding and also the optimization of laser welding parameters for creating sound weld.

1.8 OBJECTIVE AND SCOPE OF THE PROJECT WORK

The main objective of this project is to investigate and analyze the mechanical and microstructural properties of laser-welded dissimilar joints between Inconel 625 and stainless steel 316L. In order to carry out the above-mentioned research work, The following objectives are formed:

- To conduct tensile testing on the welded samples to assess the strength, ductility, and fracture behavior of the laser-welded dissimilar joints.
- To conduct hardness testing on the welded samples to Assess the effect of welding parameters on the hardness distribution across the weld zone, including the heat-affected zone (HAZ) and fusion zone.
- To conduct microstructural analysis using an optical microscope to investigate the changes in grain structure, formation of intermetallic compounds, and potential defects in the weld zone.

Based on the objectives, The scope of the present research work is detailed below:

- The project will involve experimentation with different laser welding parameters, such as power, speed, and frequency to evaluate their influence on the joint properties.
- The relationship between microstructural changes, hardness, and mechanical properties will be analyzed to assess the joint's overall performance and structural integrity.

1.9 ORGANISATION OF THE REPORT

Chapter 1 introduces about welding and types of welding techniques in general and highlighted the need and advantages of laser welding.

Chapter 2 summarizes the literature survey study which focused on the selection of the laser parameters and the various method to increases the mechanical properties of the weld

Chapter 3 describes the methodology of the study and welding, material advantages and its properties along with the experimental process.

Chapter 4 explains the fabrication procedure and types of mechanical and microstructure experimentation involved. The study of Influential parameters by Anova software

Chapter 5 discussion of tensile and microhardness results in detail along with microstructure study.

Chapter 6 summarizes the conclusion arrived from the experimental methods for laser welding of dissimilar material

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The research activities in the field of Aluminium Metal Matrix Composites (AMMCs) are done to (i) reduce the overall weight of the machinery, (ii) improve the life of machinery and equipment. AMMCs are especially used because of their low density and good corrosion resistance.

Fibre laser welding of dissimilar alloys of Ti-6Al-4V and Inconel 718 for aerospace applications

The research activities in the field of laser welding are done to (i) find the optimum parameter value for the weld, (ii) improve the life and strength of the weld. Inconel 625 and 316L are especially used because of their mechanical properties and corrosion resistance in nature.

Fibre laser welding of dissimilar alloys of Ti-6Al-4V and Inconel 718 for aerospace application. *Hui-Chi Chen & Andrew J. Pinkerton & Lin Li*

Individual effects of three processing parameters, laser power, welding speed, and laser beam offset distance from the interface, were investigated during fibre laser welding of Ti-6Al-4V to Inconel 718. Experiment results showed that welding 2 mm thick sheets of Ti-6Al-4V to Inconel 718 with an IPG 1 kW fibre laser yielded crack-free welds with a higher laser power and welding speed. Fig 2.1

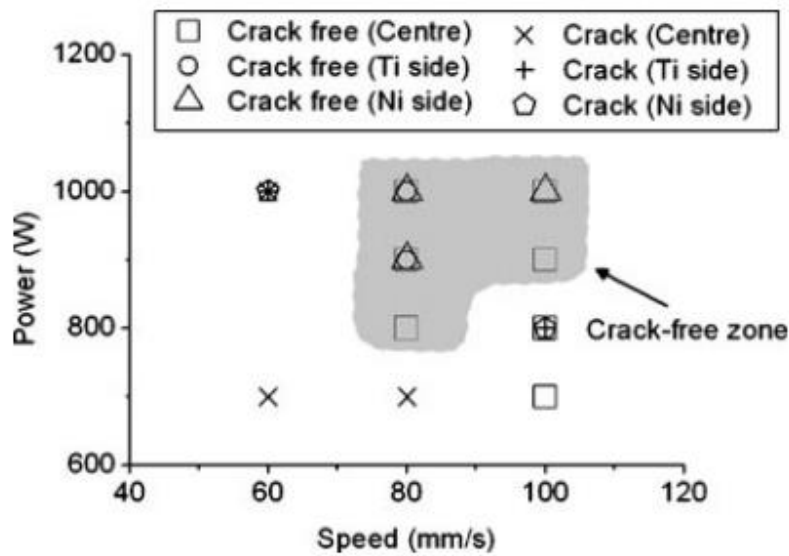


Figure 2.1 Relationship between the formation of crack, laser power, welding speed and the laser beam offset position

Offsetting the laser beam approximately 35 mm from the interface to the Inconel 718 side and using a combination of a higher laser power and a higher welding speed can produce a better quality weld with fewer hardness variations and a lower chance of cracks. This is because this method prevents the formation of Ti-Ni intermetallic brittle phases.

Joel Alexis, Jean Denis Beguin, Pablo Cerra, Yannick Balcaen. *Yb: YAG laser welding of aeronautical alloys. Materials Science Forum, 2018,*

The objective of this paper was to determine the domain of weldability by laser welding process of several aeronautical alloys: an aluminium based, commercially pure titanium, stainless steel, two nickel based superalloys, and

a cobalt base superalloy. Weld dimensions, microstructure, and mechanical properties were discussed in terms of welding parameters and materials.

To identify influential parameters of the welding process while streamlining testing, an experimental design was established with the CORICO software. The laser power, the speed welding, the focal diameter and the gas flow were the parameters studied

The welds were observed in section by optical microscopy after metallographic preparation. The mechanical properties of the welds were determined by tensile tests, carried out with a crosshead speed of 2 mm/min

The main observation was, on one side the difficulty to achieve full penetration on 1.2 mm thick sheets of AISI 321 and Inconel 625 alloys (small welding domain), and on the other side its inability to obtain sound welds on ENAW 6061 aluminum alloy.

Concerning mechanical properties, it appears that the wider is the weld, the most the UTS of the assembly is affected

Mechanical and Metallurgical Properties of CO₂ Laser Beam INCONEL 625 Welded Joints. *Harinadh Vemanaboina , Edison Gundabattini , Suresh Akella , A. C. Uma Maheshwer Rao 3 , Ramesh Kumar Buddu , Paolo Ferro ,* and Filippo Berto*

In their experimental study, dissimilar laser welding is performed between stainless steel 304 (S·S 304) and Inconel 625 alloys. The temperature is measured by changing such parameters of laser as laser power, welding speed

and nozzle distance. The parameters variations are 300–400 W, 240–480 mm/min and 0–4 mm for the laser power, welding speed and nozzle distance, respectively

Effect of laser welding parameters on the temperature distribution, microstructure and mechanical properties of dissimilar weld joint of Inconel 625 and stainless steel 304 *Li Yan a,b,, Jafar Eskandari Jam c, Mohsen Heydari Beni c, Mohammad Javad Kholoud d, Dumitru Baleanu e,f,g,, Majid Eskandari Shahraki h, Ferial Ghaemi*

In this experimental study, dissimilar laser welding is performed between stainless steel 304 (S.S 304) and Inconel 625 alloys. The temperature is measured by changing such parameters of laser as laser power, welding speed and nozzle distance. The parameters variations are 300–400 W, 240–480 mm/min and 0–4 mm for the laser power, welding speed and nozzle distance, respectively. The effect of laser power and focal length on the ultimate tensile strength of the Welding at different laser powers showed clear variation at different focal lengths and speeds of the weld. Because of the higher temperature gradient at the fusion zone, increasing the laser power from 300 to 400 W significantly improved the tensile strength of the weld joint (Fig. 9a). When the focal length was increased from 0 to 4 mm, the resulting tensile strength was reduced dramatically, by about 200 MPa, or about 100%.

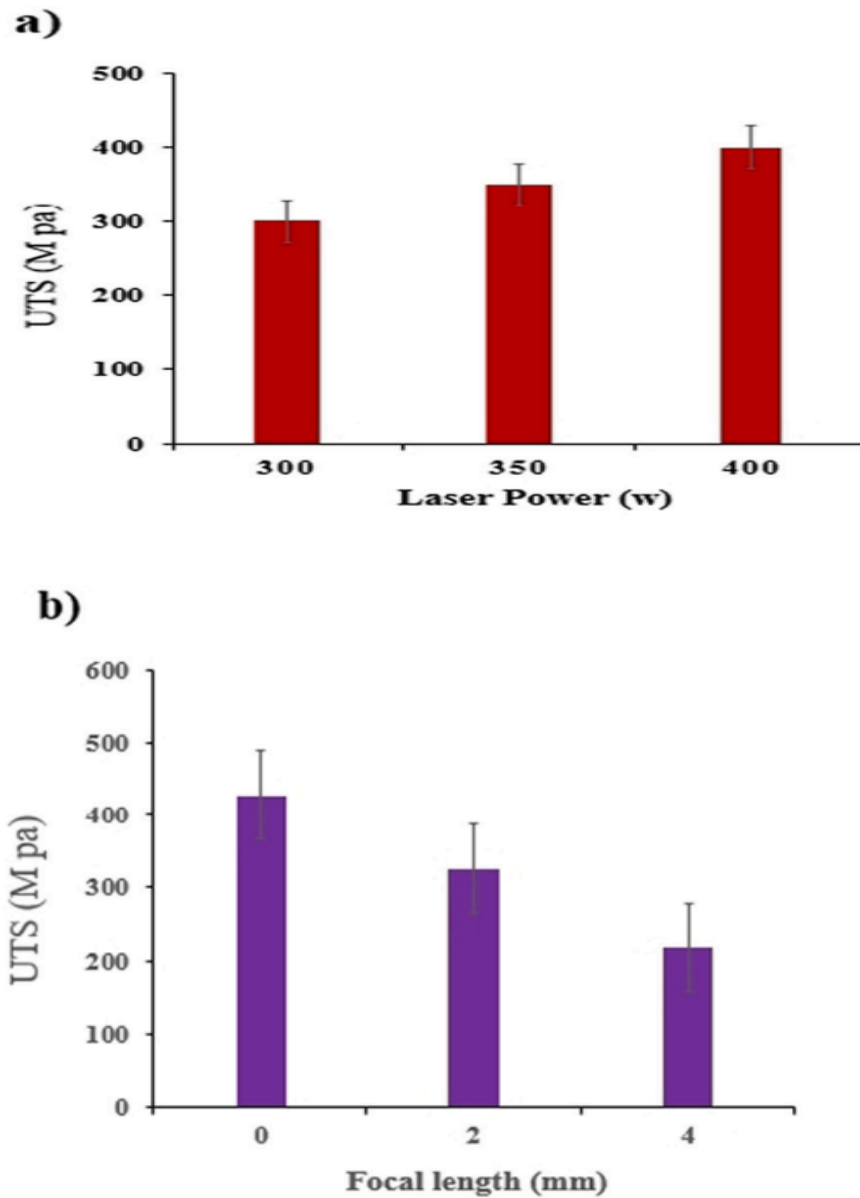


Fig. 2.2 Effect of laser welding parameters on ultimate tensile strength of the dissimilar weld joint at welding speed of 320 mm/min for a) laser power and focal length of 0 mm, b) focal length at laser power of 350 W.

The most important results of this study were Examining the laser power change effect on the temperature changes shows that the change in the laser power had a greater effect on temperature changes around the Inconel 625 side

melt pool and the change rate was about twice that of the SS 304. The tensile strength of dissimilar weld joint was significantly affected by focal length variations

2.2 SUMMARY OF THE LITERATURE REVIEW

The impact of laser welding in the industry and the welding of dissimilar materials have been the motivating factors for the extensive research work done in the literature review.

- Extensive research is being conducted on Inconel 625 due to its high strength and corrosion resistance.
- Addition of 316l plays a role in reducing the overall cost and weight of the build
- laser welding is the method chosen to weld the metal due to its cost effectiveness, simplicity and easier control.

CHAPTER 3

METHODS AND MATERIAL SELECTION

3.1 LASER WELDING

Laser welding is an advanced welding technique that utilizes a high intensity laser beam as a heat source to join materials together. It offers several advantages over conventional welding methods, making it a preferred choice in various industries such as petrochemical, aerospace industries, etc. In laser welding, a laser beam is directed onto the joint area of the materials to be welded. The laser beam moves along the joint, the molten material solidifies, creating a strong and durable weld joint. The application of laser welding has received a great attention in past few decades due to its higher productivity, higher speeds, and greater penetration as compared to other welding methods with an exception of electron beam welding.[1]

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3.1.1 SELECTION OF FIBRE LASER

Different laser sources exist such as Nd: YAG, fiber, CO₂ etc. for welding of dissimilar metals. The fiber laser produces a beam of outstanding optical quality for precise welding actions. Due to the local heat input, deep welds can be produced at high heating and cooling rates. Modern fiber laser technologies have been evaluated successfully for their ability to weld two dissimilar metals. Unlike other welding technologies, these laser welding technologies are able to be pulsed, the pulses can be shaped and hence the temperature of the weld joint can be precisely controlled at the molten joint. Specifically, the small focus diameter of fiber laser technology offers enhanced power density, a smaller

heat-affected zone, lower cycle time, and lower heat input, which can lead to a lower volume of inter-metallics and even controlled inter-metallic development

3.1.2 SHIELDING GASES FOR FIBRE LASER WELDING

Laser beam absorption and scattering due to plasma formation is of secondary importance in Fiber laser welding in contrast to CO₂ laser welding. As a result, helium and helium mixtures are not particularly useful as shielding gases in Fiber laser welding of stainless steels. The selection of shielding gases for Fibre laser welding of stainless steel is largely determined by the need to provide protection against oxidation. Argon has been used as a welding gas for low-power Fibre laser welding (less than 1000W) of small stainless steel components for many years. Argon may also be applied in high-power welding (1–5 kW) [1]. When welding austenitic steels, argon/6–10% hydrogen may be used to reduce surface oxides in order to obtain shiny weld surfaces. The selection of argon as shielding gas in turn reduces the production cost.

3.2 MATERIALS USED AND THEIR APPLICATIONS

The two materials employed in this study were Inconel 625 and stainless steel 316L. The chemical composition and application of Inconel 625 and stainless steel 316L is given below.

3.2.1 CHEMICAL COMPOSITION OF INCONEL 625 AND STAINLESS STEEL 316L

Table 3.1:Chemical composition of Inconel 625 and Stainless steel 316L

Base/Filler	C	Cr	Si	Ni	P	Mn	Nb + Ta	Mo	Fe	Other
Inconel 625	0.04	22.12	0.15	Rem.	0.012	0.09	3.38	8.32	4.48	Cu—0.05 Al—0.11 Ti—0.21 Co—0.3
AISI 316L	0.02	17.25	0.04	10.38	0.039	1.78	-	2.15	Rem.	S—0.004

3.2.2 MECHANICAL PROPERTIES OF INCONEL 625 AND STAINLESS STEEL 316L

Table 3.2:Mechanical properties of Inconel 625

DENSITY (g/cm ³)	MELTING TEMPERATURE (°C)	THERMAL CONDUCTIVITY (W/m ⁰ C)	TENSILE STRENGTH (MPa)
8.44	1350	9.8	992.9

Table 3.3:Mechanical properties of SS 316L

DENSITY (g/cm ³)	MELTING TEMPERATURE (°C)	THERMAL CONDUCTIVITY (W/m ⁰ C)	TENSILE STRENGTH (MPa)
8.027	1390	16	560

3.2.3 INCONEL 625

Inconel 625 alloy consist of a solid solution alloy of nickel, chromium and molybdenum alloy with niobium added. It is generally supplied in the annealed condition to provide utmost stability during fabrication and subsequent service. It has found to have a resistance to potable and salt or brackish water, where chloride ions cause pitting and crevice corrosion to become established. As a result, pipelines for gathering sour gas have been built with solid alloy and used in sea water heat exchangers in refineries, power plants , and offshore topside services that often need this material. The alloys properties make it an excellent choice for sea water applications, including freedom from local attack, high corrosion-fatigue strength, high tensile strength, and resistance to chloride-ion stress-corrosion cracking. The outstanding and versatile corrosion resistance of Inconel 625 alloy under a wide range temperatures and pressures is a primary reason for its wide acceptance in the chemical processing field. Because of its ease of fabrication, it is made into a variety of components for plant equipment. Inconel 625 alloy is also used extensively in the aerospace, automobile, oil extraction, oil refining, marine, waste treatment, pulp & paper and power industries.

3.2.4 STAINLESS STEEL 316L

Stainless steel 316l is proven to be more resistant to corrosion, as well as both crevice and pitting corrosion than conventional steels. The 316l offers higher creep, stronger at high temperatures, and higher stress to rupture. It is primarily used in process streams containing halides or chlorides. The low carbon content decreases carbide precipitation that sensitizes the grain boundaries of welds in these materials. The higher molybdenum content assures it will have superior pitting resistance than 304l in applications involving chloride solutions, particularly involving chloride solutions.

3.3 DISSIMILAR WELDING OF INCONEL 625 AND STAINLESS STEEL 316L

This two materials combination is used in chemical plants, aerospace, oil& gas, and nuclear applications, where high corrosion and wear resistance are essential requirements. Although these properties are superior in Inconel 625, replacing Inconel with stainless steel in strategic regions of structural components can reduce the overall costs and part weight. These two metals are used in chemical processing at very high temperatures, especially the chromic acid calcining process, which contains a diluted acid environment. The typical applications of these bimetallic combinations include the sub-scale boiler employed for the advanced stirring conversion system.[]The materials used to build this sub-scale boiler were AISI 316l for the condenser and Inconel 625 wick for the boiler.

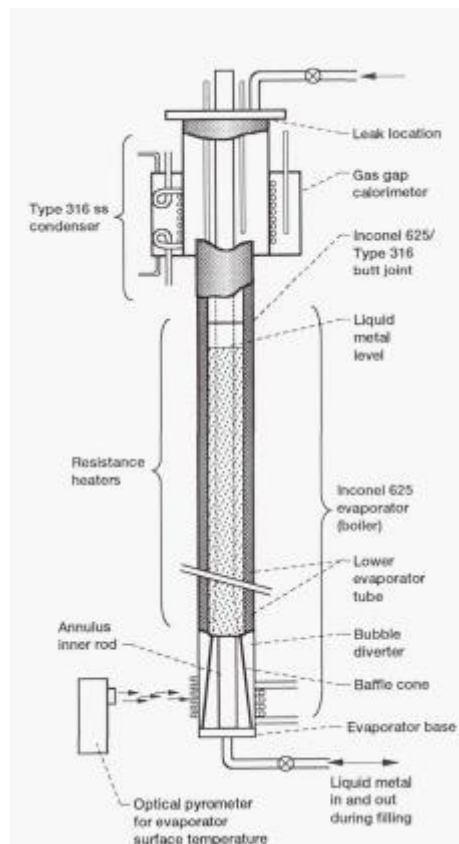


Fig3.1:NASA subscale boiler

The subscale boiler used in NASA for the stirring conversion with a butt joint of 316 on top and Inconel 625 on the bottom. This butt joint was created using arc welding which induces high heat input to both base metals which caused the micro-cracks in the Inconel side which then eventually caused failures. The major advantage of laser welding over other welding techniques is its low and controlled heat input and low distortion compared to other welding techniques, also laser welding is the most favorable in welding austenitic stainless steels because of high laser absorption by the austenitic steels. Although the combination of these metals was used in several applications, limited studies focused on these are found, so there is a need to access the mechanical and microstructural properties of this combination welded using the laser welding technique and optimization of laser parameters for the sound welds of this combination is required to use the full benefits of laser welding.

3.4 FACTORS TO BE CONSIDERED FOR WELDING DISSIMILAR METALS

3.4.1 SOLUBILITY

Solubility is the chemical property of a substance which determines its ability to dissolve in a solvent. The solubility of the two dissimilar metals must be mutual. If there is mutual solubility between the two metals, the dissimilar joints can be made successfully. If there is little or no solubility between the two metals to be joined, the weld joint will not be successful.

3.4.2 INTER-METALLIC COMPOUNDS

The problem of making welds between dissimilar metals relates to the transition zone between the metals and the inter-metallic compounds formed in this transition zone. For the fusion type welding processes, it is important to investigate the phase diagram of the two metals involved. Apriori analysis should be done about the inter-metallic compounds that will form within the transition zone between the two metals, investigating things, such as the crack sensitivity, how susceptible these are to corrosion, and its ductile ability. In some cases, it is necessary to use a third metal that is soluble with each metal in order to produce a successful joint. This analysis can be useful to decide on using a filler material or a „buttering layer“ that is easily soluble with the two dissimilar metals.

3.4.3 WELDABILITY

The level of weldability incorporating the solubility and inter-metallic compounds refers to the ability of the two metals in question to be successfully welded without resulting in cracks or any other defect. This will vary from metal to metal. Once we have determined the weldability of dissimilar metals, we can also appropriately select any other filling metals or buttering layers we shall use for a smooth transition.

3.4.4 THERMAL EXPANSION

One must also consider the coefficient of thermal expansion of the two metals involved, which relates to how these metals will change shape in response to a change in temperature. If these are too different then the internal residual stresses can greatly reduce the operating life of the newly welded metals.

3.4.5 MELTING RATES

Just as the two dissimilar metals may have different thermal expansion rates, they may also have different melting rates, which will cause an immediate problem between the two metals.

3.4.6 CORROSION

Corrosion can occur in between the transition area of the two dissimilar metals. If the two metals are on wildly different sections of the electrochemical scale, then this suggests a high level of susceptibility to corrosion, which is, of course, a damaging problem to the new weld

CHAPTER 4

EXPERIMENTAL RESULTS

4.1 INTRODUCTION

The main objective of this project research work is to study on the influential parameters of the laser welding of dissimilar materials of Inconel 625 and stainless steel 316L. The microstructure and mechanical properties of the weld depend on the parameters like laser power, speed of the weld and frequency of the weld and focal length of the lens. To investigate on the influence of variation in parameters like laser power, speed of the weld microstructure and mechanical properties of Inconel 625 and stainless steel 316L. experimental study was conducted in the following sequence:

- Selection of influential parameters of the laser welding
- Laser welding of the sample
- Tensile test
- Microhardness Test
- Microstructure analysis
- Analysis of the parameters which influenced the most by using Anova

The alloys studied are: AISI 316L stainless steel, and Inconel 625 with 2mm thickness. Their mechanical and physical properties are given in the table 1. The samples were wire cut and butt welded with IPG Fiber laser machine

4.2 EXPERIMENTAL PROCEDURE

The candidate metals used in the studies were cold-rolled Inconel 625 plates and AISI 316L rectangular plates. Wire-cut Electric discharge machining using brass wire was used for fabricating the samples to the required dimensions of 100mm x 100mm x 2mm before welding. The plates were cleaned using acetone so as to make the surfaces free from any contaminants. Autogenous weld bead tracks were carried out using a 3kw Fibre laser welding machine with different welding speeds for different sample combinations. The Argon gas was used for shielding in order to prevent the weld puddle from contamination or oxidation. The welding was carried out in keyhole mode. The weldments were polished using abrasive sheets of siC and welds were examined using Daksh metallurgical microscope. Vickers microhardness tester was employed to obtain the hardness profile of the welded coupons. The coupon was tested with a load of 100gf and a dwell time of 10s for the load to act upon. the specimens from the welded joints were extracted and shaped according to ASTM E8M-21 standards, to perform the transverse tension test using Instron universal testing machine (UTM).

4.2.1 LASER WELDING SETUP

The machine has a maximum continuous power of 3 kilowatts

1. IPG Fiber laser

- 3KW power
- 3-axis movement along with a rotatory motor

2. Monitor

3. Software used (CAD DSP)

4. argon shielding gas

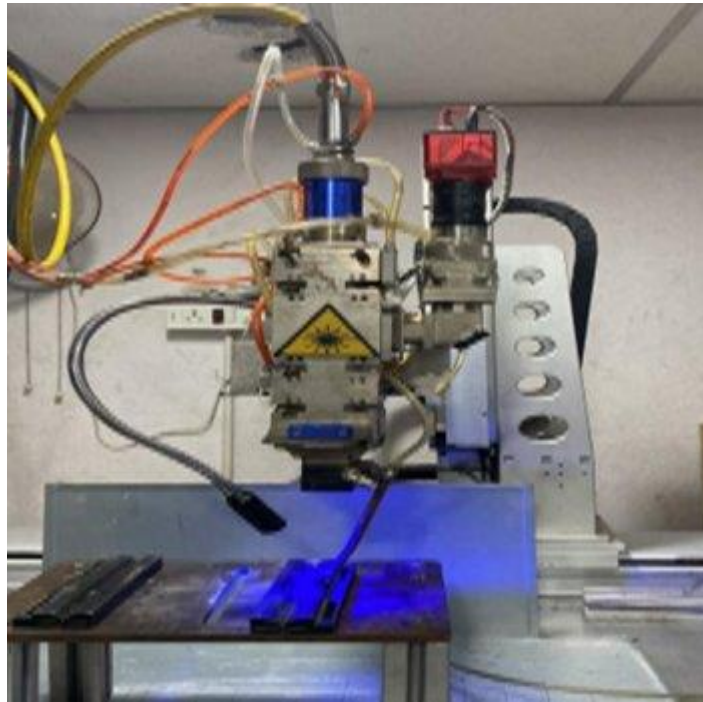


Fig. 4.1 Laser welding machine with setup of the IPG Fibre laser machine

4.3 SELECTION OF LASER WELDING FREQUENCY

To identify influential parameters of the welding process while streamlining testing, an experimental design was established to study the optimum frequency for the welding. Low frequency is selected because it has less spatter in welding. In research papers low frequency less than 75Hz is used by conducting trial runs we found the optimum frequency is 45 Hz

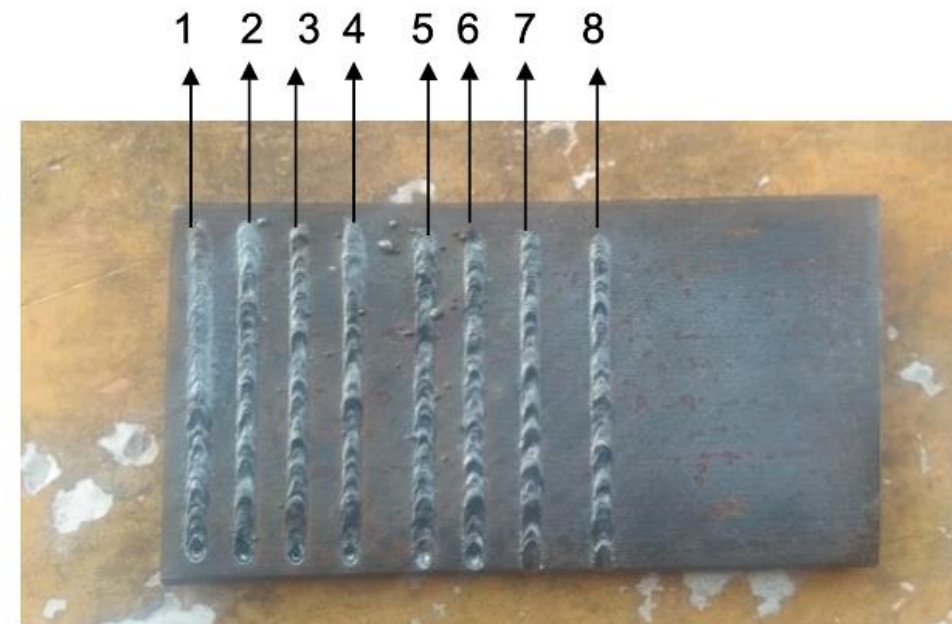


Fig. 4.2 Laser weld with different frequencies in Hz (5 Hz, 15 Hz, 25 Hz, 75 Hz, 500 Hz, 1000 Hz, 3000 Hz, 5000 Hz) from left to right

4.4 TAGUCHI APPROACH FOR DOE

Taguchi method is used widely in the design of experiments which provide a simple, efficient, and systematic approach for the optimization of experiments. And it gives improved performance, higher quality of weld product, and reduced cost of conducting the experiment. For the purpose of designing the laser welding using the Taguchi method, key parameters were identified and their variation was chosen based on the literature review discussed in Chapter 2.

Table 4.1: Parameters-wise matrix levels used for the DoE

Sample No	Laser Power (W)	Welding Speed (mm/s)
1	1500	6
2	1500	9
3	1500	12
4	2100	6
5	2100	9
6	2100	12
7	2700	6
8	2700	9
9	2700	12

4.5 WELD SAMPLES PENETRATION:

From fig.3.2 and 3.3 we can observe that full penetration of the weld was seen in the samples from 1 to 5 this indicates that the weld speed significantly reduces the penetration of the weld. Lower welding speeds lead to a more complete fusion due to more time given for colling and a better bonding between the weld and base metal, resulting in a stronger overall joint

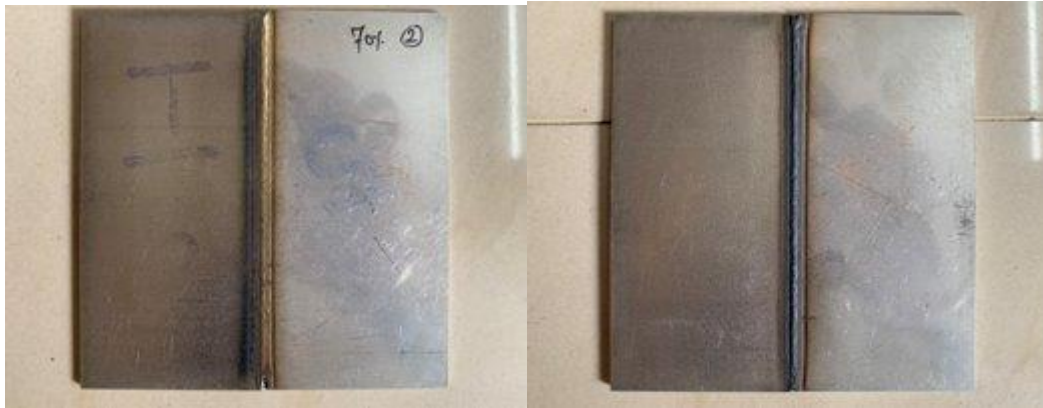


Fig. 4.3 laser weld of 5th sample front and back side view

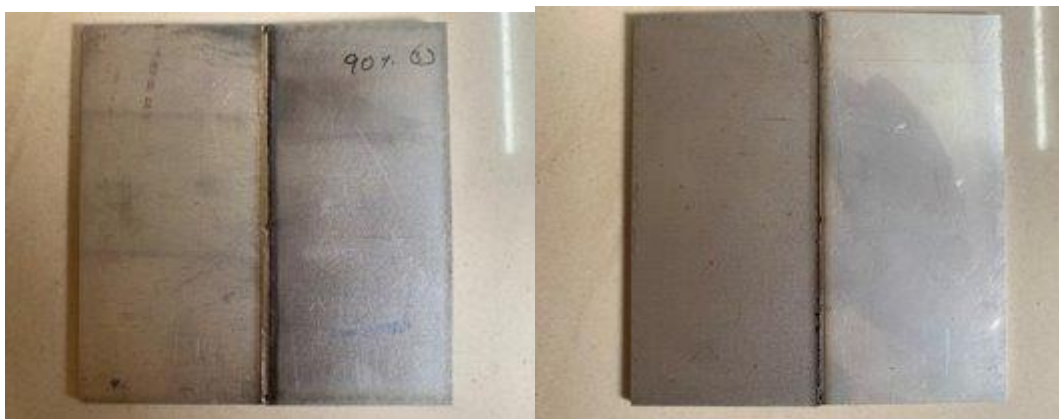


Fig. 4.4 Laser weld of 9th sample front and back side view

4.6 TENSILE TEST:

The welds were cut using electro discharge wire cut machine and various specimens were extracted such as tensile specimens, and microstructural specimens from 3 different locations across the weld plate. The specimen size was as per ASTM standards using electro discharge wire cut machine.

From each welded plate, two tensile specimens were prepared as per ASTM E8M - 21 standard

4.6.1 TENSILE SPECIMEN

The tensile test specimens were carved out of a casting plate with cross-sectional dimensions of 100 mm x 20 mm x 2mm. The specimens were cut out by ASTM E8.

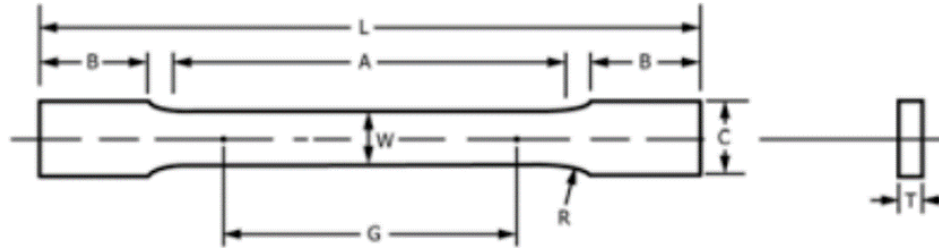


Figure 4.5: ASTM E8 Specimen

Nomenclature of the ASTM E8 Specimen

Where,

$A = 60\text{mm}$, $B = 45\text{mm}$, $C = 20\text{mm}$

$L = 170\text{mm}$, $R = 13\text{mm}$, $G = 50\text{mm}$

$W = 12.5\text{mm}$, $T = 2\text{ mm}$

The specimens were cut out of the produced casting plates according to ASTM standard E8. Tensile tests often are used to predict the behaviour of a material under forms of loading other than uniaxial tension. The strength of a material often is the primary concern.

The strength of interest may be measured in terms of either the stress necessary to cause appreciable plastic deformation or the maximum stress that the material can withstand. These measures of strength are used, with appropriate caution (in the form of safety factors), in engineering design. Also of interest is the material's ductility, which is a measure of how much it can be deformed

before it fractures. Rarely is ductility incorporated directly in design; rather, it is included in material specifications to ensure quality and toughness. Low ductility in a tensile test often is accompanied by low resistance to fracture under other forms of loading. Elastic properties also may be of interest, but special techniques must be used to measure these properties during tensile testing.

4.6.2 UNIVERSAL TESTING MACHINE

The most basic kind of mechanical test you can run on a material is called a tensile test, often known as a tension test. You can rapidly determine how a material will respond to tension forces applied to it by pulling on it. You can determine the material's strength and how much it will elongate when you apply load on it.

The tensile strength and compressive strength of materials are tested using a universal testing machine (UTM), also known as a universal tester.

Specifications:

- Capacity = 10 to 100 KN
- Voltage = 440V
- Automation grade = Automatic

Load is applied progressively when the specimen is secured in the universal testing machine. After a particular amount of time, the specimen began to elongate. The neck was created after a length of time. The specimen then began to shatter. The elongation percentage, yield load, and ultimate load are all indicated.

- i. the tensile properties such as yield strength, ultimate tensile strength were computed from computerized universal testing machine.

The tensile stress-strain curves of the nine SS316L/IN625 joints were plotted. According to the true stress-strain curve, all test samples except TS3 and TS9 has obvious plastic deformation before fracture, indicating that the fracture mechanism of the joint is ductile fracture [14]. In comparison with the other test samples, Test sample 5 displayed the highest tensile strength (664MPa) with promising yield strength (388MPa). The TS3 and TS9 showed brittle fractures with elongation percentages of less than 20 in both test samples. It is useful to compare the mechanical properties of the dissimilar joints to those of components combining Inconel and stainless steel in reported literature using different manufacturing methods [15,16,17,18]. The comparison was depicted in the Fig 5.2 in terms of the ultimate tensile strength vs elongation. It is important to note that the failure location in the TS 5 was on the stainless side, as seen in the reported results [16,19]. The fracture location was limited to the stainless steel side indicates that Inconel and stainless steel have a good bonding strength. In TS9, separation of elements and the formation of laves phases encouraged a reduction in the tensile strength of the weld joint. It was reported that the presence of laves phases facilitates the nucleation of and propagation of the micro-cracks and so it is damaging on the strength [20].

4.2: TENSILE RESULTS:

Table 4.2: Tensile Test report of the weld

S.No.	Speed (mm/s)	Power (KW)	Yield Strength (N/mm ² or Mpa)	Ultimate Tensile Strength (N/mm ² or Mpa)	Fracture Location
1	3	1500	374	654	Base
2	3	1500	372	656	Base
3	3	1500	361	658	Base
4	6	1500	374	653	Base

5	6	1500	380	652	Base
6	6	1500	373	641	Base
7	9	1500	368	470	Weld
8	9	1500	363	433	Weld
9	9	1500	350	486	Weld
10	3	2100	366	642	Base
11	3	2100	364	652	Base
12	3	2100	358	657	Base
13	6	2100	391	665	Base
14	6	2100	388	662	Base
15	6	2100	385	665	Base
16	9	2100	367	660	Base
17	9	2100	386	652	Base
18	9	2100	378	656	Base
19	3	2700	372	652	Base
20	3	2700	379	646	Base
21	3	2700	375	644	Base
22	6	2700	383	658	Base
23	6	2700	434	657	Base
24	6	2700	412	661	Base
25	9	2700	384	474	Weld
26	9	2700	404	509	Weld
27	9	2700	406	607	Weld

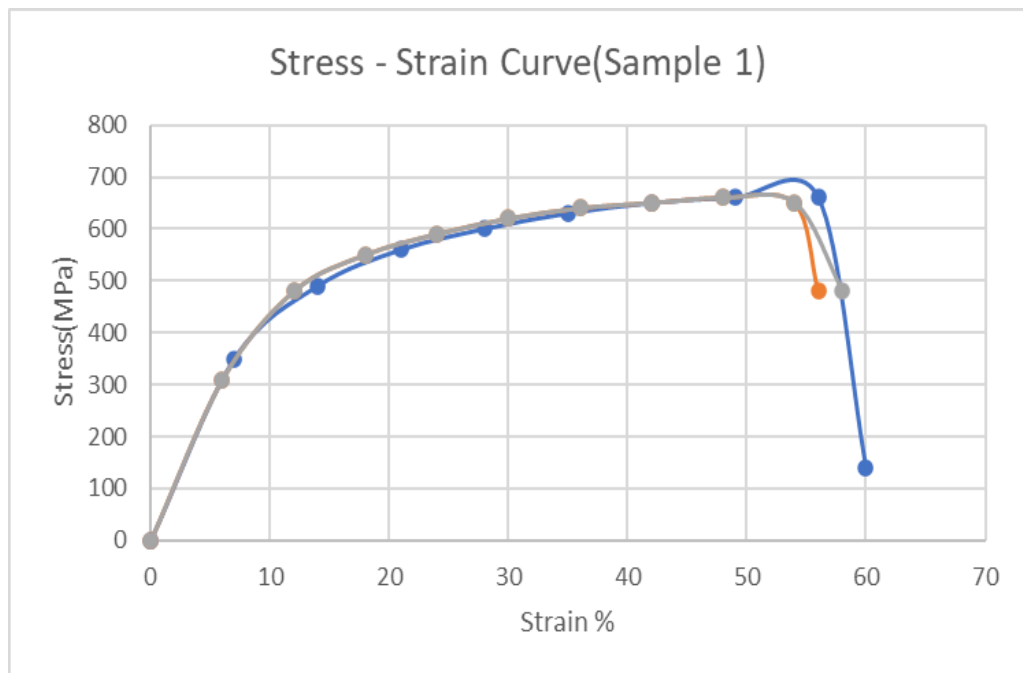


Fig4.6: Stress Strain curve for 1500 W and 6 mm/s welding speed

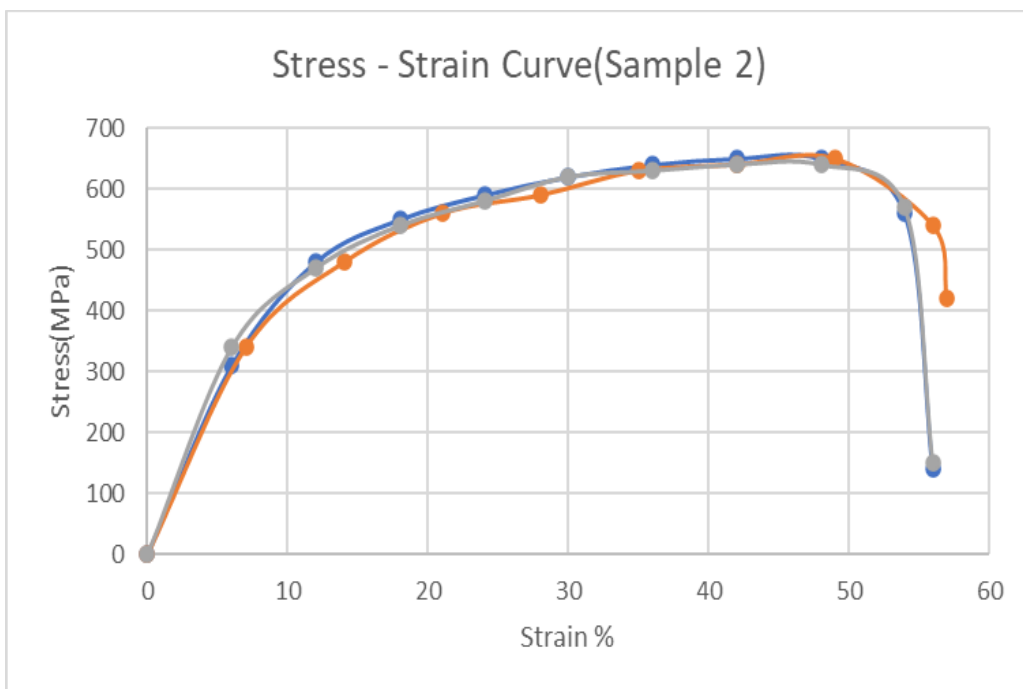


Fig4.7: Stress Strain curve for 1500 W and 9 mm/s welding speed

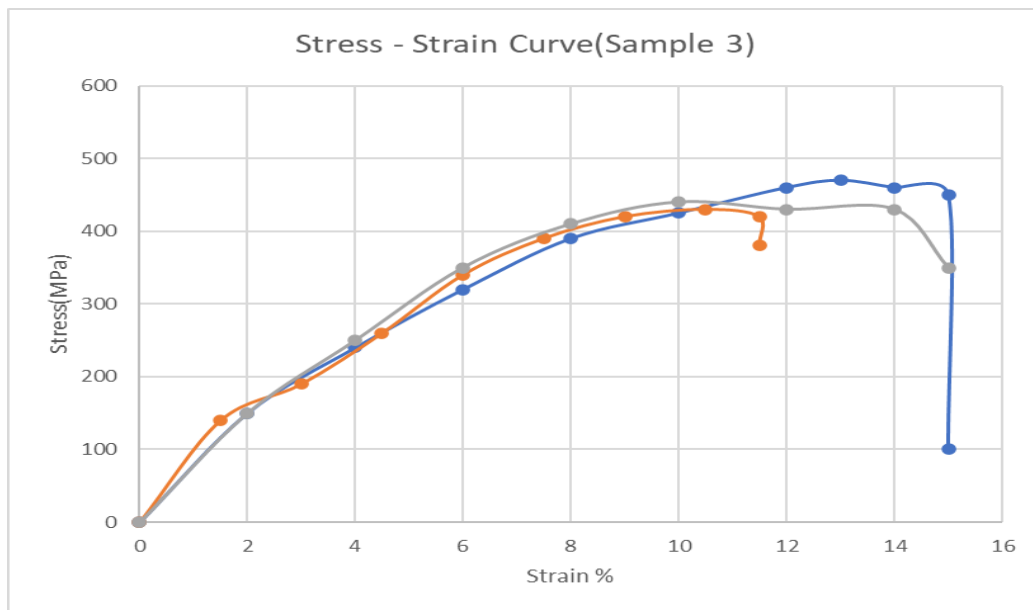


Fig4.8: Stress Strain curve for 1500 W and 12 mm/s welding speed

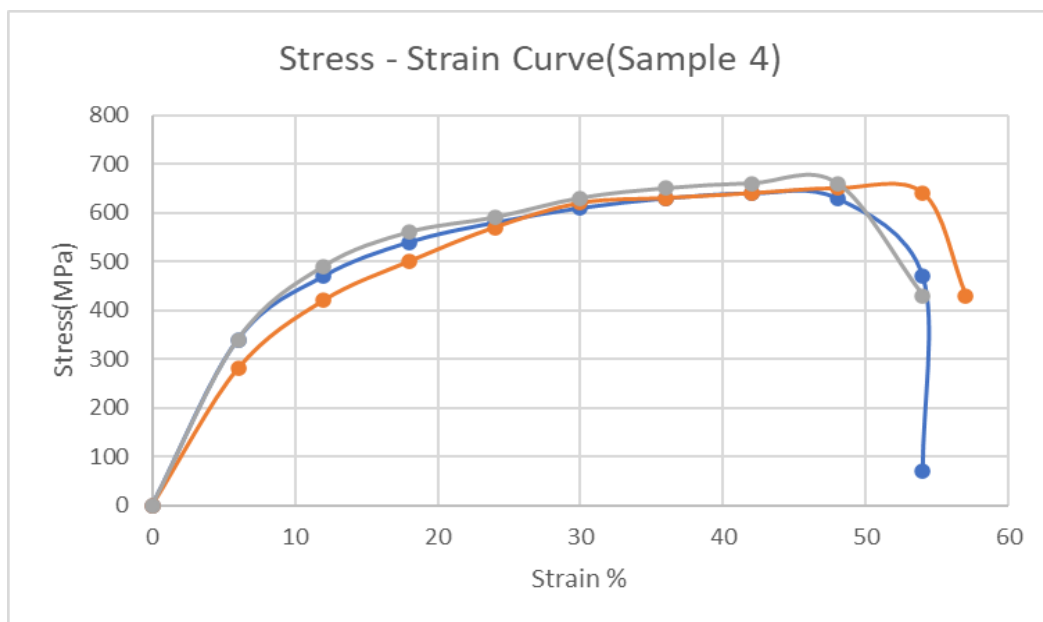


Fig4.9: Stress Strain curve for 2100 W and 6 mm/s welding speed

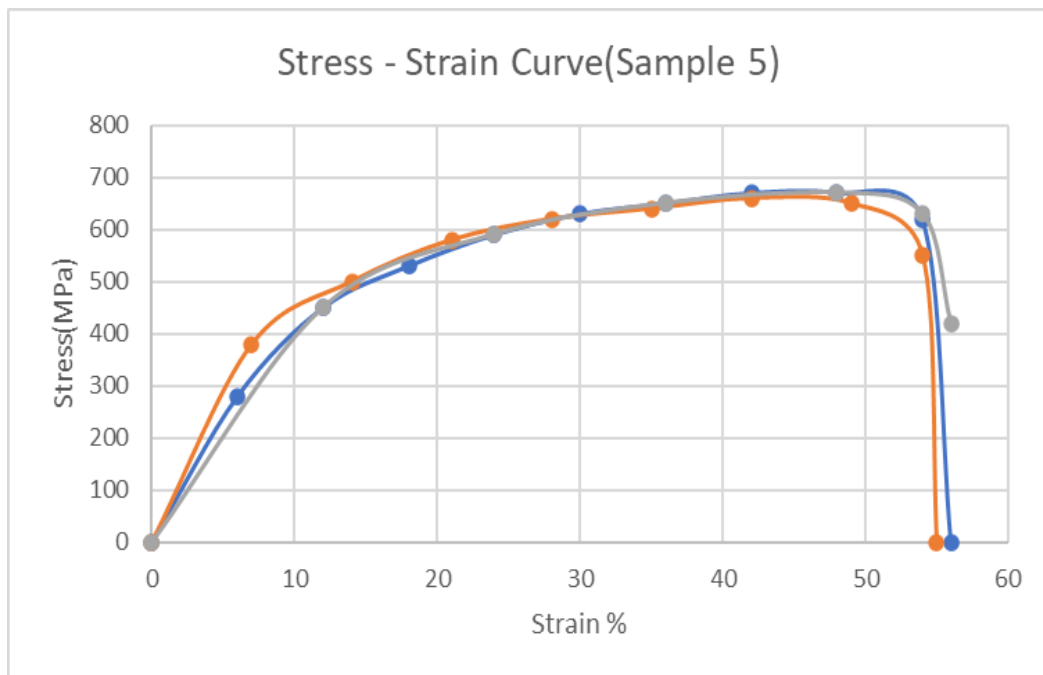


Fig4. 10: Stress Strain curve for 2100 W and 9 mm/s welding speed

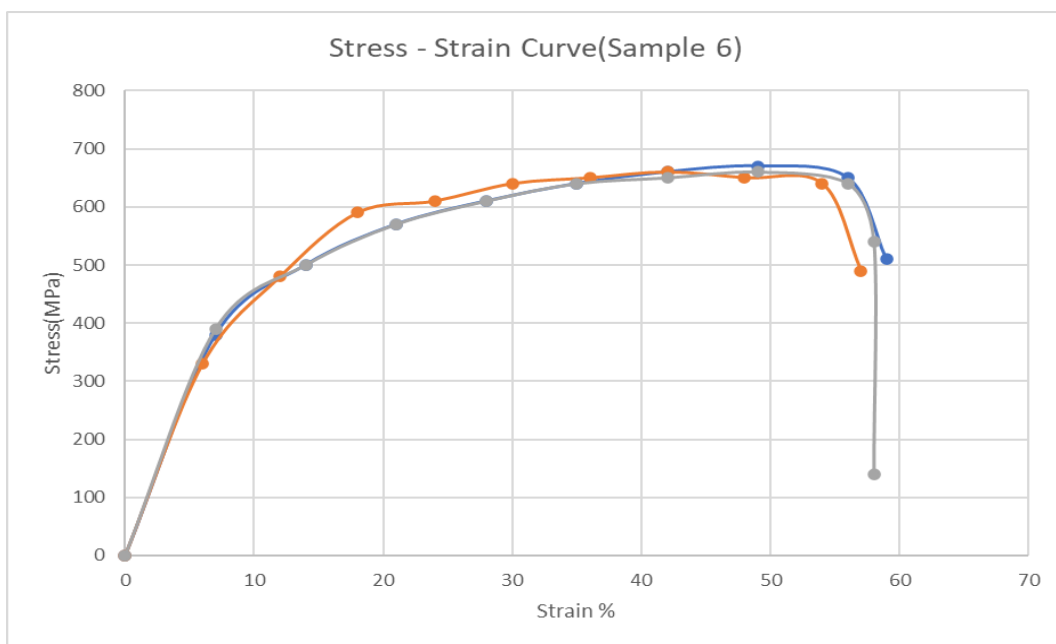


Fig4.11: Stress Strain curve for 2100 W and 12 mm/s welding speed

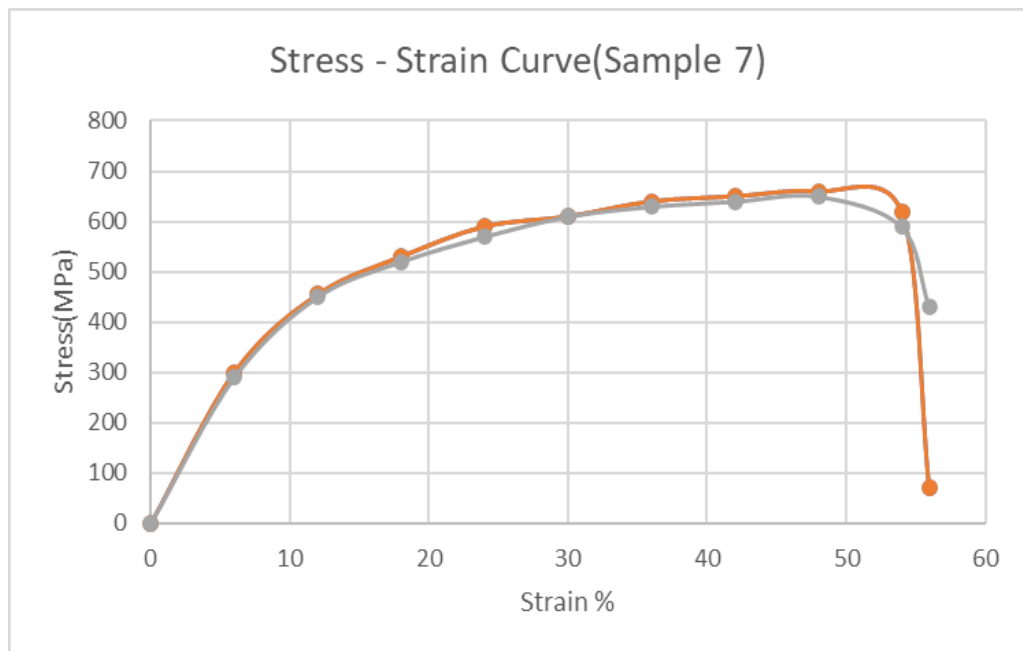


Fig4.12: Stress Strain curve for 2700 W and 6 mm/s welding speed

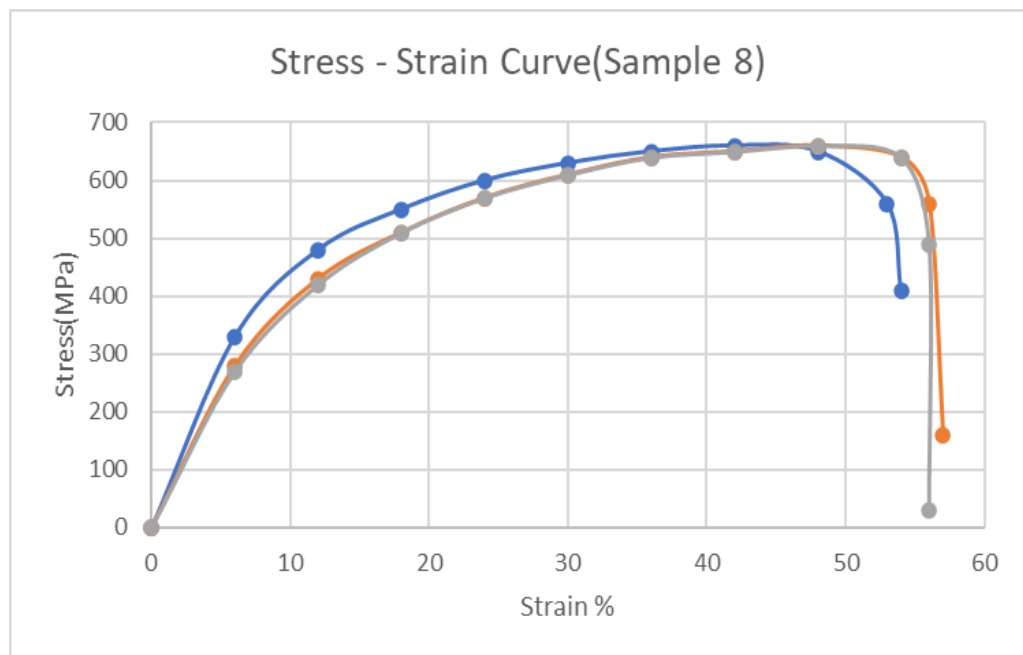


Fig4.13: Stress Strain curve for 2700 W and 9 mm/s welding speed

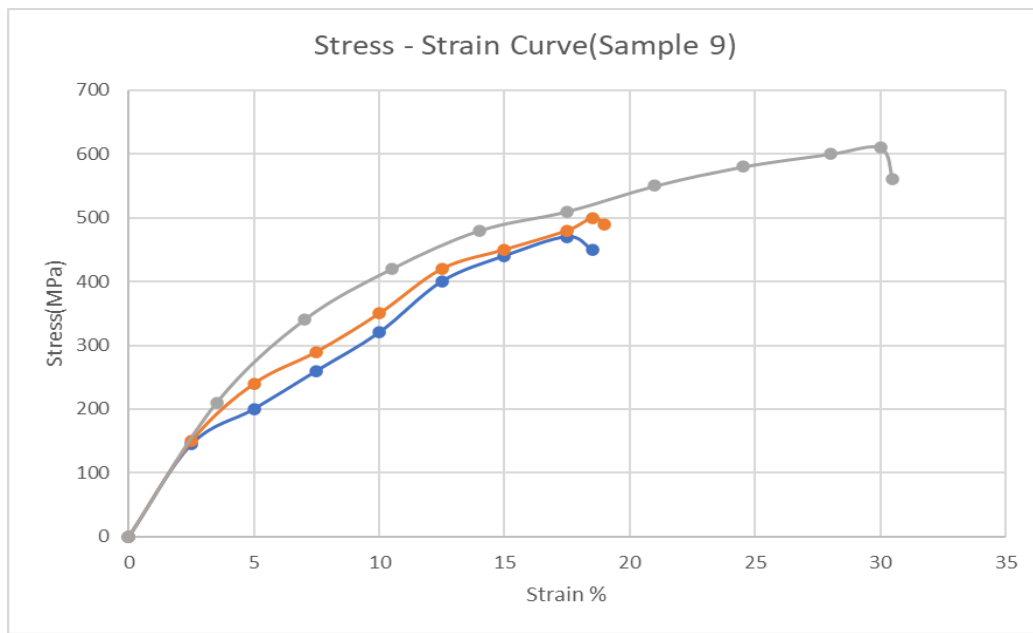


Fig4.14: Stress Strain curve for 2700 W and 9 mm/s welding speed

4.7 MICROHARDNESS

Using a Vickers microhardness testing machine, a microhardness investigation was performed on flat weld metal samples across the joints in the middle of the thickness direction according to ASTM E384 standard. A 100kgf load was applied for 10 seconds with a diamond indenter to determine the microhardness across the weld cross-section in the transverse direction. Figure 3.8 depicts a pictorial representation of a Vickers microhardness testing machine. The diagonals of the indentations were measured using the microscope reticle, and the Vickers hardness number was calculated for each indentation using the formula. The hardness value for each zone is taken in 3 different places at the same zone and the average is taken for accuracy. The microhardness testing was performed at three locations on each sample, including The heat-affected zone (HAZ) at the SS 316L side, The HAZ at the Inconel 625 side, and The fusion zone (FZ). The microhardness tests were taken with 1.5mm intervals from the weld center on both sides.



Figure 4.15 Daksh Microhardness Tester Model: MHB1A

Table 4.3 : Hardness values of nine samples across the weld

S.No	BM	HAZ(ss)	HAZ(ss)	FZ	HAZ(In)	HAZ(In)	BM
1	205.8	178.1	196.9	273.01	416.1	441.9	365.1
2	168.3	187.2	282.3	272.3	361.3	313.3	370.2
3	175.2	190.6	233.9	222.8	365.3	411.2	378.7
4	188.7	193.7	234.9	321.7	362.2	313.3	358.2
5	178.5	217.2	286.9	291.9	423.9	336.2	361.6
6	166.4	183.7	192.1	250.5	344.4	342.5	381.4
7	164.6	188.3	202.6	264.9	370.4	390	355.3
8	200.3	197.8	206.2	251.5	326.5	487.6	352.5
9	184.4	174.2	235.5	210.7	390.8	455.4	360.5

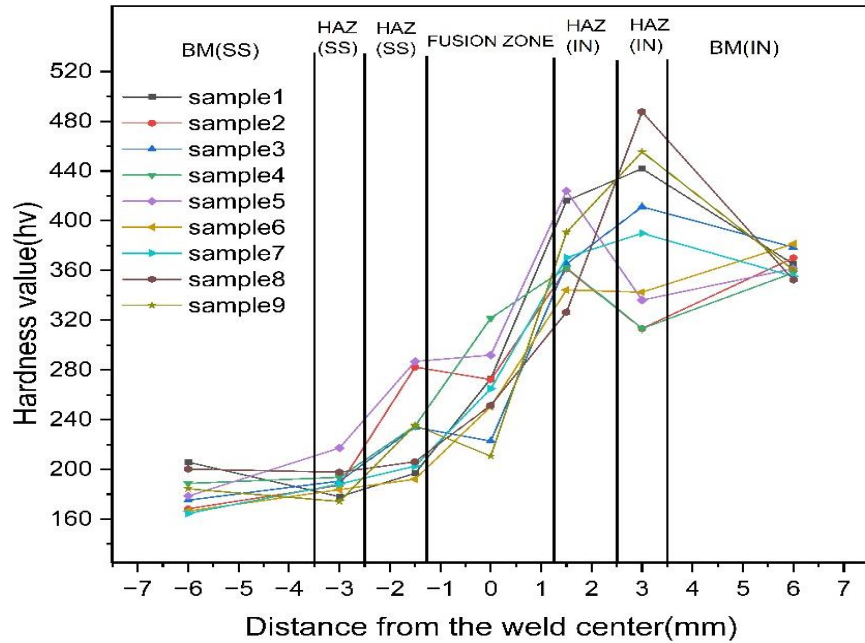


Fig4.16: Hardness profile across the weld for all nine combinations

The figure shows the microhardness profile for all nine dissimilar joint combinations combined. As shown in the figure, the micro-hardness of all nine dissimilar combinations exhibits an increasing trend from stainless steel 316L to Inconel 625. The average micro-hardness of the weld metals for all combinations is 262.1HV. This shows that the weld bead is stronger than the weaker base metal stainless steel. The average micro-hardness of base metal of stainless steel 316L and Inconel 625 are 181.4HV and 364.8HV. Test Sample 4 which is welded with a combination of laser power 2100W and a welding speed of 6mm/s has the highest micro-hardness at the fusion zone of 321.7HV. Test sample 9 which is welded with a combination of laser power of 2700W and welding speed of 12mm/s has the lowest micro-hardness at the fusion zone of 210.7HV. The difference between the highest and lowest value of hardness is 110.7HV which is a high value. This shows that the laser parameters greatly affect the Hardness of the fusion zone. The hardness values at the heat-affected zones near the fusion zone of stainless steel for all samples increase from the base metal. There was a steep increase in the hardness value from the fusion zone to the heat-affected zone of Inconel 625.

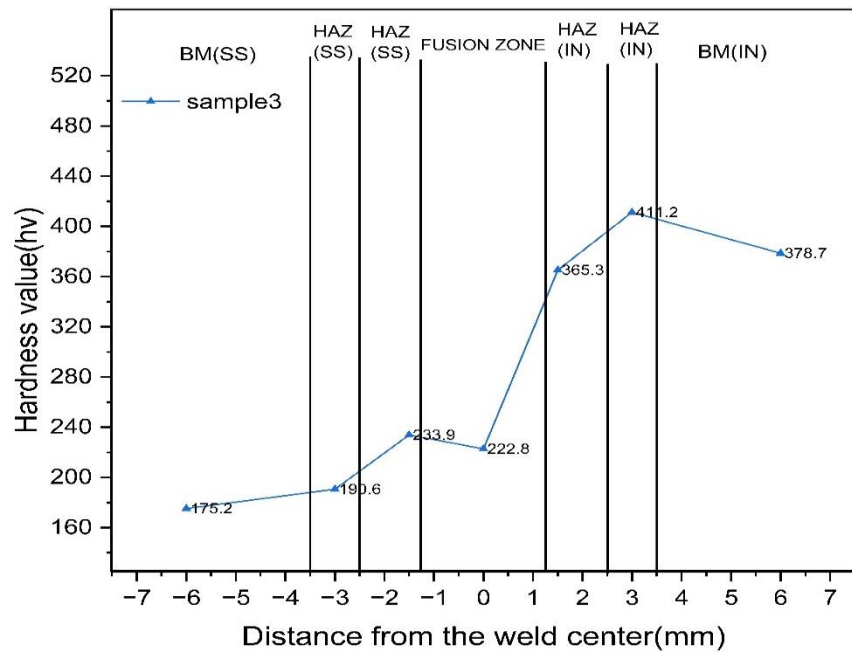


Fig4.17:a)Hardness profile of test sample 3

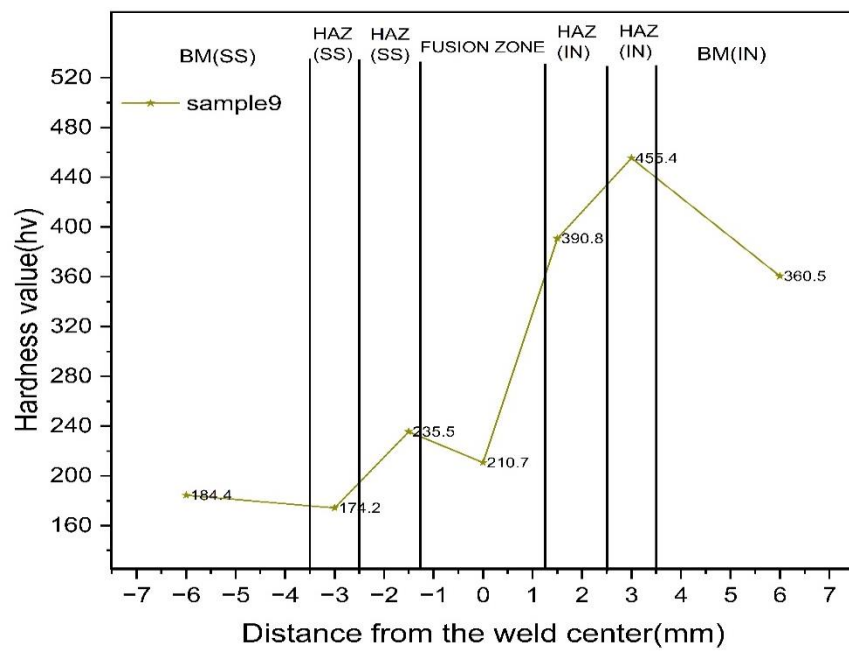


Fig4.17:b)Hardness profile of test sample 9

The figure and figure show the hardness profile of sample 3 and sample 9 respectively. From the figure, it is evident that there is sudden decrease in hardness from heat affected zone of stainless steel to the fusion zone and again the hardness value starts to increase. This trend was shown in the samples which were failed at the weld during tensile test. There is a significant softening in the HAZ-FZ boundary of the sample 3 and sample 9.

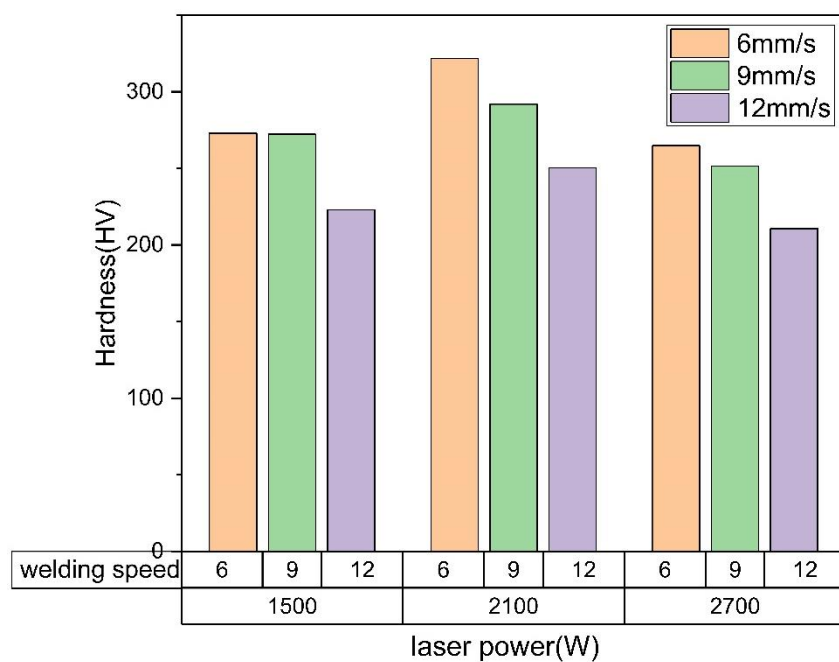


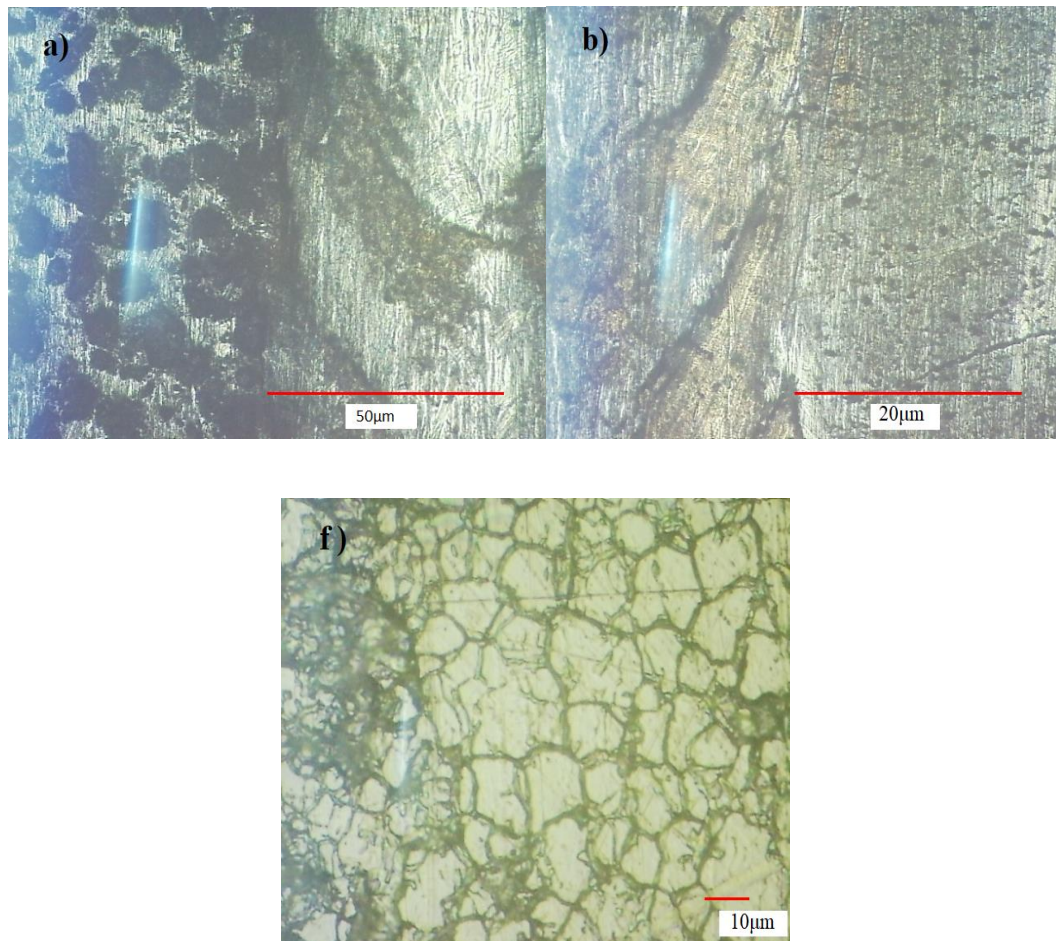
Fig4.18: Hardness value vs welding speed and laser power

The figure shows the graph for the hardness value of different speeds at the same laser power. From the graph, it is clear that for all three-laser power, the hardness value decreases with increasing speed at the fusion zone.

4.8 MICROSTRUCTURAL ANALYSIS

he microstructure Specimens were cut from the welded sample using wire electrical discharge machining and followed by the standard mechanical

polishing. The grain structure was photographed by the Daksh metallographic optical microscope. The optical micrographs of the different regions in the laser-welded optimal joint were taken for the purpose of analysis. Weld metal, fusion zone, heat affected zone boundary,



**Fig 4.19 :Sample 5,a)weld interface between Inconel and fusion zone,
b)weld interface between stainless steel and fusion zone,f)stainless steel
base metal**

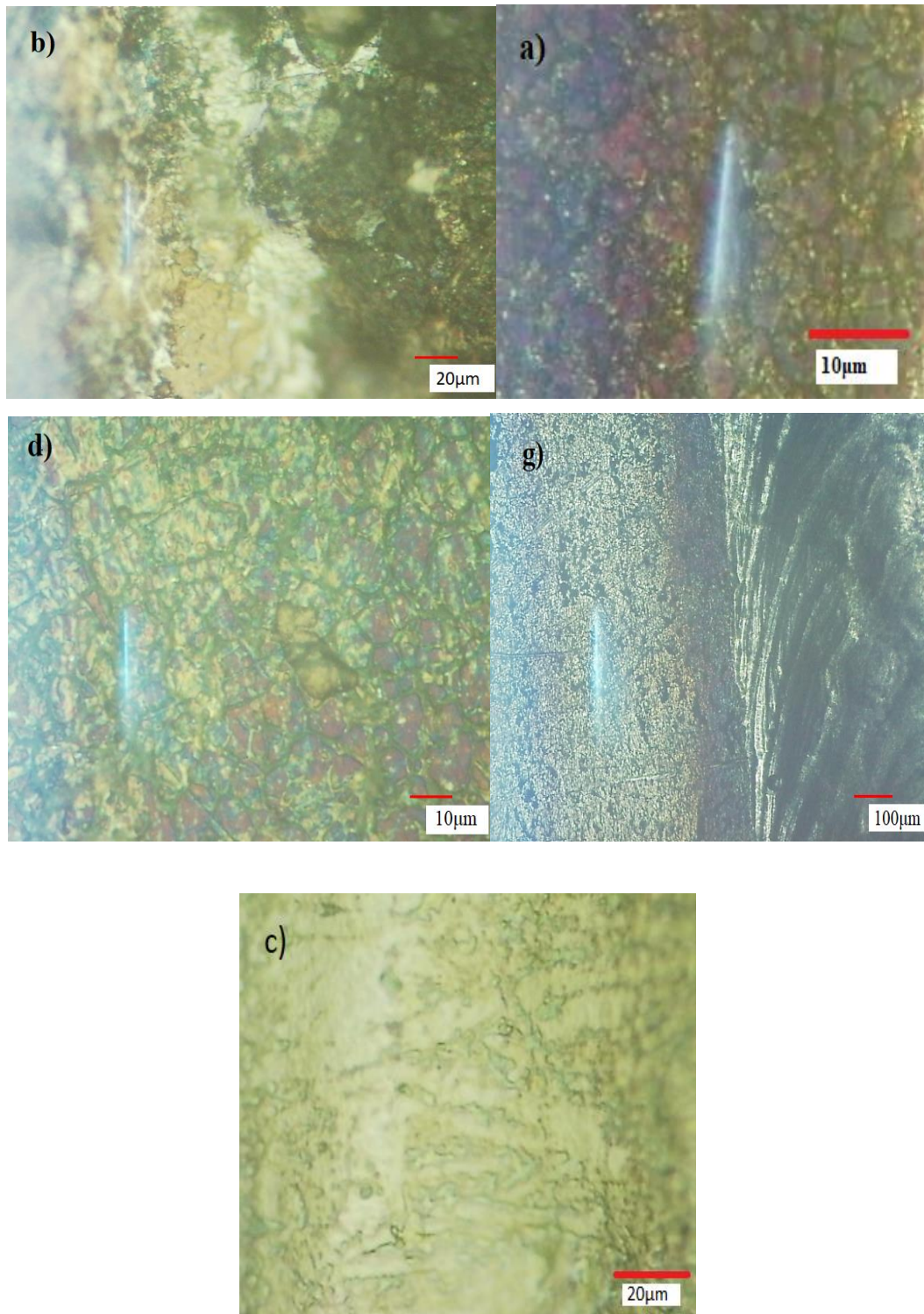


Fig 4.20 : Sample 9, a) weld interface between stainless steel and fusion zone, b)1000x of weld interface between Inconel and fusion zone, c)fusion zone near stainless steel, d) HAZ of stainless steel, g)50x of weld interface between stainless steel and fusion zone

The base metal structure of sample 5 which showed the highest tensile strength, and sample 9 which showed the lowest tensile strength are shown respectively. The laves phase formation, shrinkage voids and martensite formation at the stainless steel can be seen from the micrographs. The micrographs were analyzed by correlating with the previous studies conducted on Inconel/stainless steel dissimilar joints. It revealed the cause of the brittle failure at the weld interface. The microstructure supports the results from tensile and hardness.

4.9 ANOVA

Analysis of Variance is a collection of statistical models used in order to analyze the differences between group means and their associated procedures (such as “variation” among and between groups).

ANOVA is done using Minitab software. ANOVA is done to find the significance of input parameters and their interaction parameters. This is done by comparing the obtained F-values with table F-values. When the obtained F-values are greater than the table values, the parameter corresponding to it is found to be significant in affecting the hole quality character.

4.9.1 TERMINOLOGIES IN ANOVA

Variance:

variance is the measure of the variability in the Ultimate Tensile Strength. ANOVA analyses the total variance in the data and breaks it down into different

components related to laser welding parameters, specifically laser power and laser weld speed.

Between-Parameter Variation:

It indicates whether the choice of laser power or weld speed significantly influences the performance.

Within-Parameter Variation:

within-parameter variation as the variability that arises from differences within each laser power or weld speed setting.

Degrees of Freedom:

Degrees of freedom is the number of independent pieces of information used to estimate a parameter or calculate a statistic for the laser welding parameters.

F-Test:

The F-test is a statistical test used in ANOVA to determine whether there are significant differences between the means of different groups or treatments. F-test assesses whether there are significant differences in UTS based on varying laser power and laser weld speed.

F-Table:

The F-table is a reference table that provides critical values for the F-statistic at different levels of significance and degrees of freedom. It is used in conjunction with the F-test to determine the statistical significance of the observed F-statistic.

By comparing the calculated F-statistic to the critical value from the F-table, you can determine whether the observed differences in UTS is among the laser power or weld speeds. If the calculated F-statistic exceeds the critical value, it suggests that the differences are significant and you can reject the null hypothesis in favor of the alternative hypothesis.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 INTRODUCTION

The following section presents a comprehensive analysis of the results obtained. The experimental methodology involved laser welding of dissimilar metal samples followed by a series of tests, including tensile, hardness, and microscopy examinations. The subsequent discussion provides an in-depth interpretation of the obtained results, highlighting the effects of welding parameters and material compatibility on the mechanical behavior and microstructural evolution of the welded joints.

5.2 OPTIMAL PARAMETER SELECTION

The parameters identified for obtaining a sound laser weld are (i) Penetration, (ii) Intermetallic compounds, and (iii) Cooling rates of the weld. To identify the best combination of parameters from nine different combinations, the weld samples are subjected to types of analysis (i) Mechanical properties analysis to identify the strongest weld combination and (ii) microstructural analysis- To investigate the precipitate formation and change in dendritic structure. The analysis of the laser weldments to identify the optimized parameters is discussed in the subsequent sections.

5.2. MECHANICAL PROPERTIES

5.2.1 TENSILE TEST

From the tensile results the Speed vs Ultimate Tensile Strength bar graph is plotted which gives comparison of UTS of different laser welding speed and for same power. Where the power represented in percentage form the maximum 3KW. From the graph we can conclude that the laser welding speed affects the UTS in greater effect. The UTS decreases drastically in the welding speed of 12 mm/s. The welding speed of 9 mm/s gives maximum UTS strength in all the power section

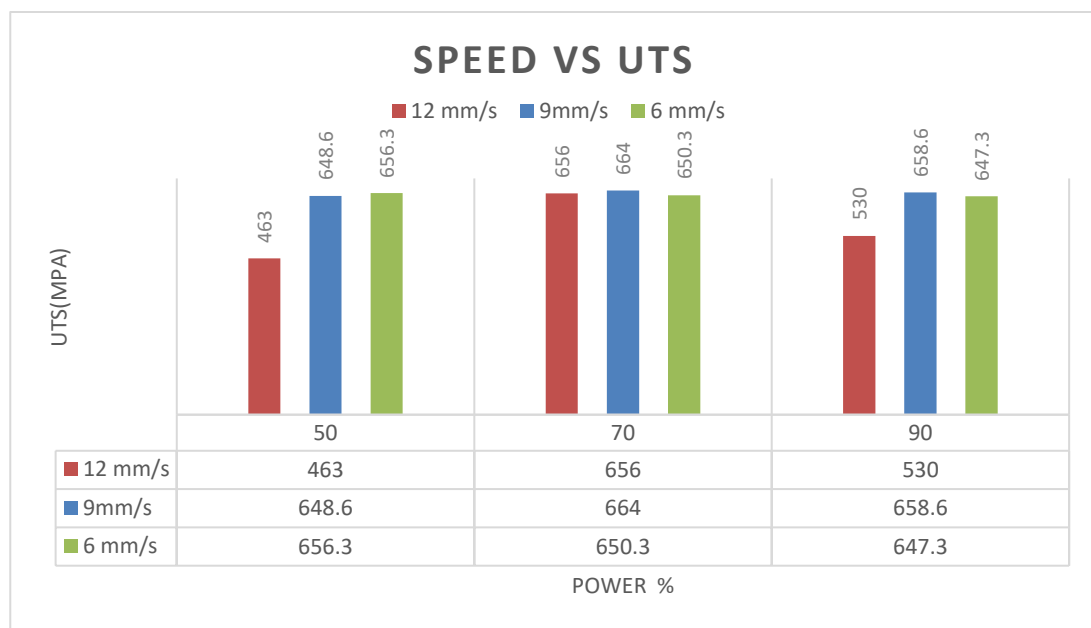


Fig 5.1: Welding Speed vs Ultimate Tensile Strength of the nine test samples

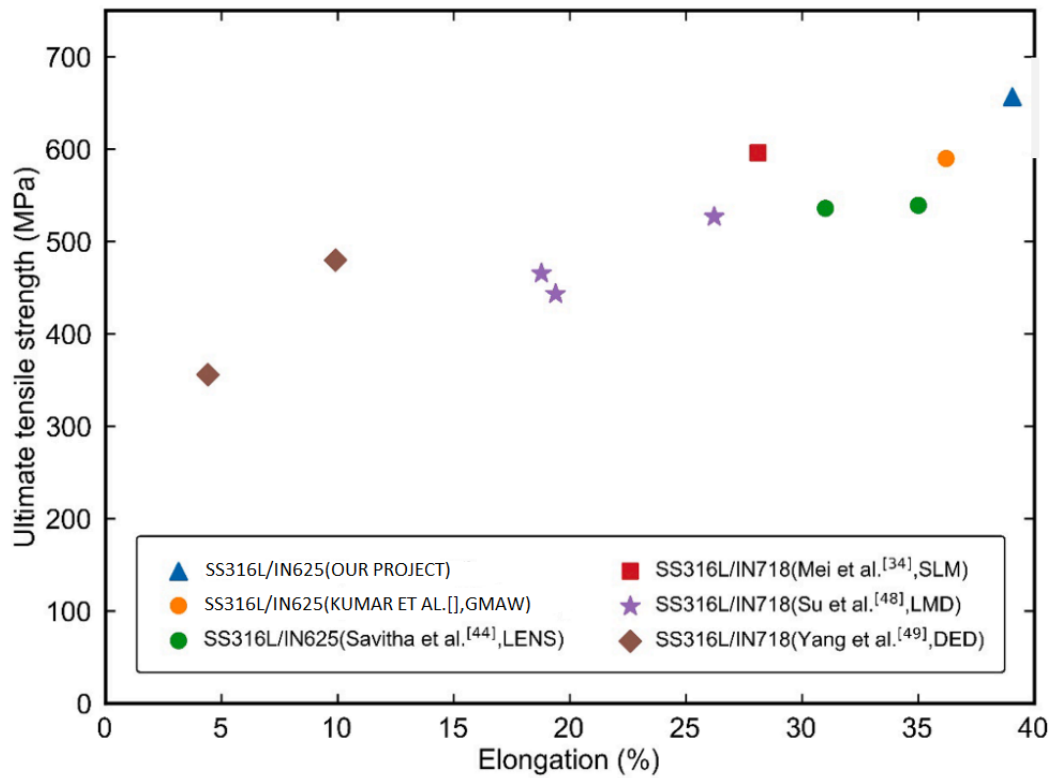


Fig 5.2 : Comparison of maximum Ultimate tensile strength of Inconel/stainless steel joints obtained from different studies

5.2.2 MICROHARDNESS TEST

It is inferred from the graph that there was a steep increase in the hardness value from the fusion zone to the heat-affected zone of the Inconel. This is because of the higher attentiveness of Nb in the fusion zone from the Inconel due to the distortion developed in the matrix by this element owing to its large radii [3]. There is also an increase in hardness at the Heat-affected zone of Inconel due to the migration of Cr and Ni from the Inconel side to the location near the fusion [3]. The peak hardness of 487.6HV was experienced adjacent to the weld interface of Inconel, which could be reasoned for the formation of carbon-enriched martensite due to the chain-like structure.[4]Sample 3 and test sample 9 failed at the weld interface of stainless steel. It can be found that there are black lath martensite parallel to the weld bead In both the 3rd and 9th test samples [4]. Also, the difference between the

hardness value of HAZ and the fusion zone indicates this fracture. The reason for the higher hardness of the HAZ of 316L is due to the higher coefficient of heat conductivity compared to Inconel[21]. The coarse-grained martensite associates with an increase in the fault energy and dislocation density which tends to improve the hardness. The hardness values are in line with the tensile test results as it shows a decrease of hardness in the fusion zone from the HAZ zone of the stainless steel.

5.3 MICROSTRUCTURAL ANALYSIS

The optical micrographs were analysed to identify the reason for the brittle fracture at the interface between fusion zone and stainless steel. The figure:5.1f shows the base metal structure of stainless steel, which is predominantly of equiaxed austenitic grain. This grain structure has also been reported before [5]. There does not appear to be any carbide precipitates in the microstructure of stainless steel. This is due to proper annealing [6]. The figure:5.1a and figure:5.1b show the heat-affected zone adjacent to the weld interface of stainless steel 316l and Inconel 625 respectively for sample 5. It shows there is a smooth transition from base metal to weld in the stainless steel and there is carbon precipitates formation in the Inconel side. The figure:5.1d shows the fusion zone and base metal interface of Stainless steel, it is evident that there is a columnar dendritic structure near the fusion zone on sample 5. There is no formation of carbon precipitates in the stainless steel. There are also columnar structures near the Inconel fusion zone if seen at 1000x. There is a mix of columnar dendritic structure and equiaxed structure in the fusion zone. The solidification begins at the interface of the weld and base metals and progresses toward the center of the molten pool. Due to this, the solidification time at the lateral sides of the weld pool is shorter compared to the center of the weld, which results in finer grains at the lateral sides of the weld metal. The formation

of columnar-dendritic structure at the interface of weld and base metals is because of the higher gradient temperature [2].

It can be seen that the fusion zone near the stainless steel of sample 9 has a finer dendritic microstructure. This is due to higher welding speed [2]. Figures 5.2 b and 5.2d show the presence of shrinkage voids. The shrinkage voids were produced because of the lack of material flow into the space between connected dendrites during solidification [6]. The thermal conductivity of stainless steel 316L ($16 \text{ W/m}^\circ\text{C}$) is greater than that of Inconel 625 ($9.8 \text{ W/m}^\circ\text{C}$). Consequently, this causes high conduction of heat to the stainless steel side. The phenomenon of micro segregation of elements strongly depends on the heat input and cooling rate. This high heat conduction to stainless steel results in the segregation of elements. It is usually found that segregation of Nb, Mo, and Ti occurred during weld solidification near the stainless steel side[7]. The yellowish spots in between the dendrites in the figure:5.2a are laves phases, which can be due to the segregation of the above elements. Ahmad et al. [8] reported the segregation of Mo and Nb during the laser butt welding of Inconel 625 and stainless steel 2205. The samples with low welding speed have high tensile strength, so low cooling rate, results in proper diffusion into the weld metal. This can be seen in sample 5. Also, black lath martensite is formed on the HAZ of stainless steel due to a high cooling rate. This occurs in low-carbon steels like 316L[2].

5.4 ANOVA

Analysis Of Variance is done to study the influence of laser welding parameters on Ultimate Tensile Strength of the welded material.

Table 5.1 : Analysis of Variance for UTS

Source	DF	Seq SS	Adj SS	Adj MS	F _{test}	F _{table}	P _{contribution} (%)
power	2	21263	21263	10631	16.97	6.94	15.74
speed	2	65678	65678	32839	52.41	6.94	48.63
power* speed	4	36831	36831	9208	14.69	7.71	27.27
Error	18	11279	11279	627			8.35
Total	26	135050					100.00

S = 25.0318 R-Sq = 91.65% R-Sq(adj) = 87.94%

INFERENCE

From the table we can infer that F_{test} values are greater than F_{table} values for power, speed and their combination. Hence all three are significant factors. The highly influential factor on the UTS is the speed with contribution of 48% followed by combination of speed and power at 27.27 % and power at 15.74%. The contribution of residual error is 8.35%.

Main effects Plot

Main effects plot: Shows the relationship between the output parameter and every single input parameter separately.

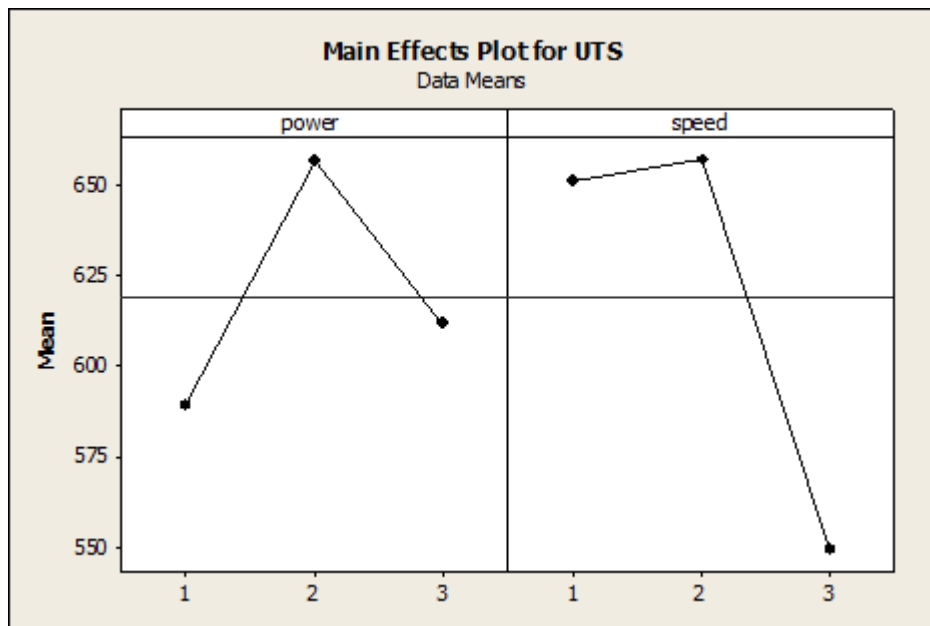


Fig 5.3: Main Effects plots for UTS

From the main effect plots, the optimal combination of power and speed is level 2 and level 2 respectively which represents the welding power of 2100 W and speed of 9mm/s respectively. To obtain high ultimate tensile strength we should select the optimum laser power of 2100 W and laser welding speed of 9mm/s.

CHAPTER 6

CONCLUSION

6.1 CONCLUSIVE SUMMARY

An attempt was made in this research work to study the effect of laser welding and laser welding parameters on mechanical and microstructural properties of dissimilar welded joints of Inconel 625 and stainless steel 316L. Nine different combinations of laser parameters were chosen to identify the most optimal parameter to weld these two austenitic metals for the present study. Best combination was chosen based on penetration of weld, high Tensile strength, Optimal hardness, less formation of laves phases and statistical results from ANOVA. The welding was successfully performed and the following results were obtained:

- During the solidification of molten metal, the morphology of the weld structure changed from columnar dendrite at the interface of the weld joint and base metals to cellular dendritic at the center of the weld metal due to a gradual decrease of the temperature gradient.
- In the tensile test, sample 3 with a laser power of 1500W and welding speed of 12mm/s, and sample 9 with a laser power of 2700W and welding speed of 12mm/s failed at the weld interface between stainless steel and the Fusion zone. The failure of weld at the sample 3 was because of inadequate penetration.
- The martensite lath formation at the stainless steel HAZ of sample 9 was the main reason for failure at the weld interface and the presence of laves phase within the fracture surface of sample 9 encouraged the formation of micro-cracks.

- The maximum value of hardness was obtained at the HAZ of Inconel 625(487.6HV) due to the formation of black carbon precipitates. The increase in the welding speed caused the decrease in hardness value in the fusion zone due to the formation of more fine grains.
- Formation of martensite grains caused the increase in hardness value of HAZ of stainless steel in sample 9 due to the high cooling rate.
- The tensile strength depends on both welding speed and tensile strength and the optimal parameter that showed the highest tensile strength of 664MPa with excellent penetration was sample 5 which was welded with laser power of 2100W and a welding speed of 6mm/s. Its hardness value at the fusion zone was 291.9HV.
- From the main effect plots, the optimal combination of power and speed is 2100 W and 9 mm/s. respectively.
- Finally, It should be stated that low input of heat and high welding speed caused inadequate weld penetration, on the other hand, high input of heat and high welding speed caused the formation of martensite grains and laves phases. This causes a brittle fracture at the weld interface.

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